UDS
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AVENGER 5

long-range, high-performance heavy loitering munition UAV platform

USER MANUAL

Model	AVENGER 5 V1
Version	UDS-A5UM-001
Release Date	2026-08-07

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1 Introduction

1.1 Scope of the Document

The purpose of this user manual is to provide information about the **Avenger 5** loitering munition platform and accompanying **SwarmC2** command and control system.

The document covers:

- Technical information about the platform.
- Preparation of the Avenger 5 loitering munition for flight.
- Operational procedures of the Avenger 5 UAV.
- Using the SwarmC2 application for mission planning and flight.

WARNING: Read Before Operation
This document must be used together with all applicable user manuals and supporting documentation covering system safety, handling, storage, transportation, preparation, operation, and associated equipment. Do not handle, prepare, power on, or operate the system or any associated equipment unless you have read and understood the applicable documentation and received the required training and authorization.

1.2 General Features

A long-range, high-performance heavy loitering munition UAV, engineered for high electronic warfare resilience and offering a modular warhead payload option. Designed for versatility and accuracy, the **Avenger 5** seamlessly integrates with various explosive payloads. After target acquisition, it autonomously executes its mission without reliance on ongoing radio communication or GNSS signals, delivering powerful and precise strikes, even in complex battlefield conditions.

- EW resiliency (frequency hopping, dead-reckoning in GNSS-denied environments).
- Fully automatic guidance system using gimbaled sensor (EO/IR).
- Auto-targeting using visual tracking or geolocation reference.
- Low visual, acoustic, thermal, and radar signature.
- Integration with SitaWare HQ, ATAK, DELTA.
- Pneumatic catapult launch system.
- Option for recoverability.

NOTE: Vehicle capabilities strongly depend on the concept of operations during use. This document provides nominal characteristics measured under standard conditions.

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1.3 General Overview

Key physical components of the Avenger 5 UAV:

- EO/IR gimballed seeker.
- Fuselage and payload bay.
- Wings (Left/Right upper and lower panels).
- Ruddervators.
- Foldable propeller.
- Antennas.

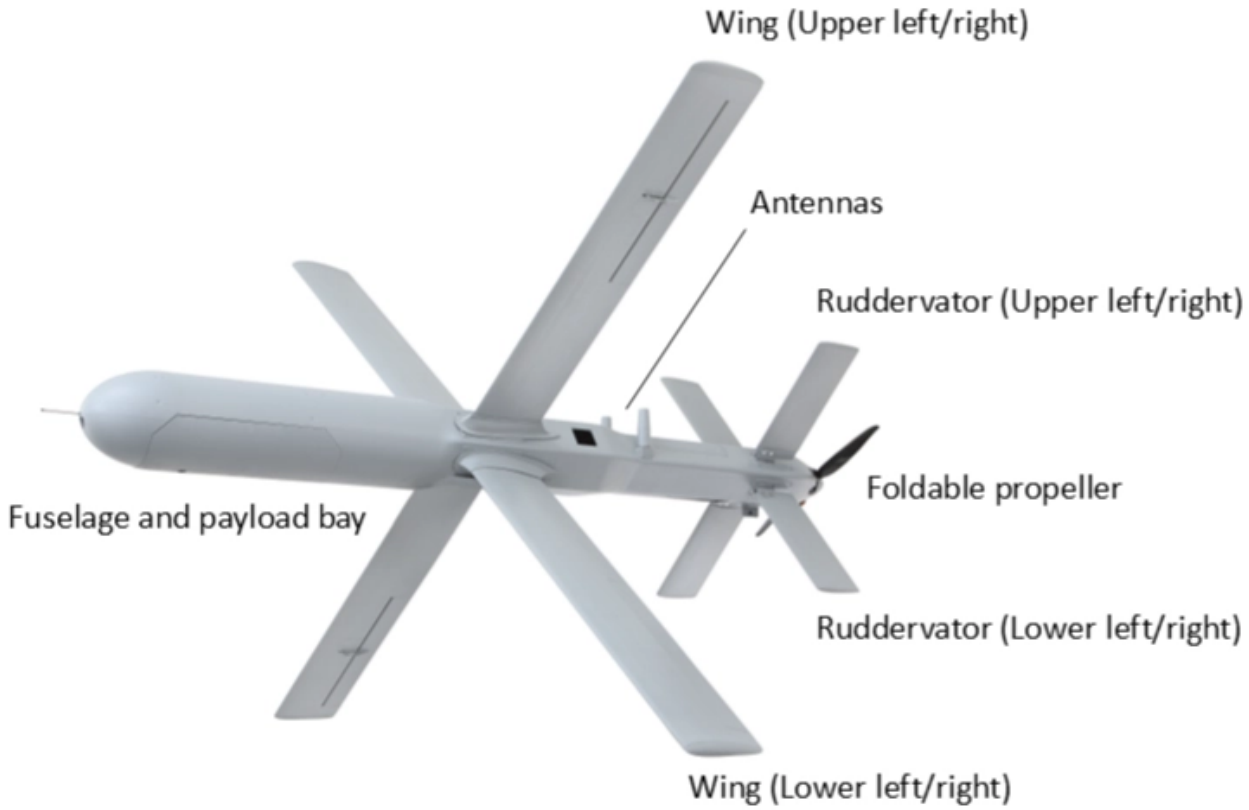


Figure 1.1: Avenger 5 physical component map

NOTE: Placement of particular components might differ according to specific configuration of the product.

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1.4 Mechanical Dimensions

Table 1.1: Mechanical dimensions

Parameter	Value
Wingspan	1.8 m
Assembled UAV length	1.6 m
Assembled UAV height	1.1 m

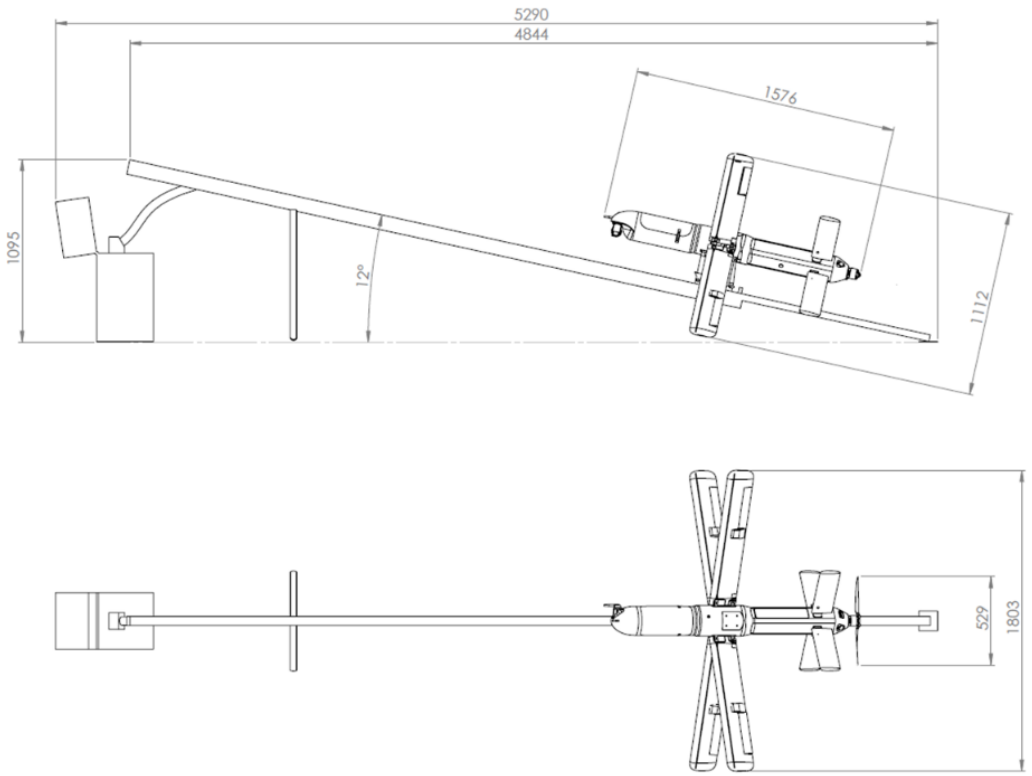


Figure 1.2: General launcher and UAV dimensions

1.5 Weight and Balance

Table 1.2: Weight and balance

Parameter	Value
Maximum take-off weight	17 kg

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1.6 Flight Characteristics

Table 1.3: Flight characteristics

Parameter	Value
Cruise speed	30 m/s
Max horizontal speed	38 m/s
Max dive speed	55 m/s
Max dive angle	22 deg
Min speed	26 m/s
Flight time	90 min
Launch type	Catapult/Tube
Recovery system	No
Command and control range (RLOS, depending on configuration)	60 km / 180 km

1.7 Seeker Characteristics

Table 1.4: Seeker Characteristics

Parameter	Value
Type	Gimballed EO/IR sensor.
Zoom	X40 (X20 + X2 digital)
Day camera	1280x720 (HD)
Thermal camera	1280x720 (HD)
Pitch	-45° to $+135^{o}$
Yaw/Roll	-165° to $+165^{o}$

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1.8 Product Contents

Table 1.5: Avenger 5 UAV

#	Component	Qty.	Comment
1	Fuselage assembly with all hatches closed and vertical stabilizer attached. Battery included	1	-
2	Wing section left	2	-
3	Wing section right	2	-
4	Ruddervator	4	-
5	RBF flag	2	-

Table 1.6: Ground Control System (GCS)

#	Component	Qty.	Comment
1	Radio BOX	1	-
2	Ground BOX	1	-
3	Rotator	1	-
4	Gitzo tripod with head	1	-
5	GCS battery	2	-
6	Panasonic FZ-G2 tablet	2	-
7	Panasonic FZ-G2 tablet battery	2	-
8	MARS antenna 23 dBi	1	-
9	Power cable with ODU connector	2	3 m
10	Ethernet cable	1	15 m
11	Antenna cable with N-type connector	2	0.25 m
12	LAN cable with ODU connector	2	3 m

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Table 1.7: Spare parts and service equipment package

#	Component	Qty.	Comment
1	Panasonic FZ-G2 Tablet battery charging station	1	-
2	GCS and UAV battery charger	1	-
3	Spare attachment bolts	30	-
4	Propeller blades	2	-
5	Propeller assembly	1	-
6	Set of tools	1	-
7	Panasonic FZ-G2 tablet spare batteries	2	-

Table 1.8: Transportation and storage cases

#	Component	Qty.	Comment
1	Case for Tactical backpack Avenger 5 UAV	1	-
2	Case for Tactical backpack GCS	1	-
3	Case for Spare parts and service equipment	1	-

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2 Overview

The **SwarmC2** system consists of three main components:

Application
: the operator software, running as a web application on one or more tablets. It provides the map view, mission planning, flight controls, and payload interface.

Ground Station
: the ground-based system deployed in the field. It interfaces with operator devices, UAVs, radio equipment, and external services, and acts as the central hub for data routing, video streaming, storage, and coordination.

UAV
: the unmanned aerial vehicle that receives commands, transmits live telemetry, and streams payload data back to the ground station.

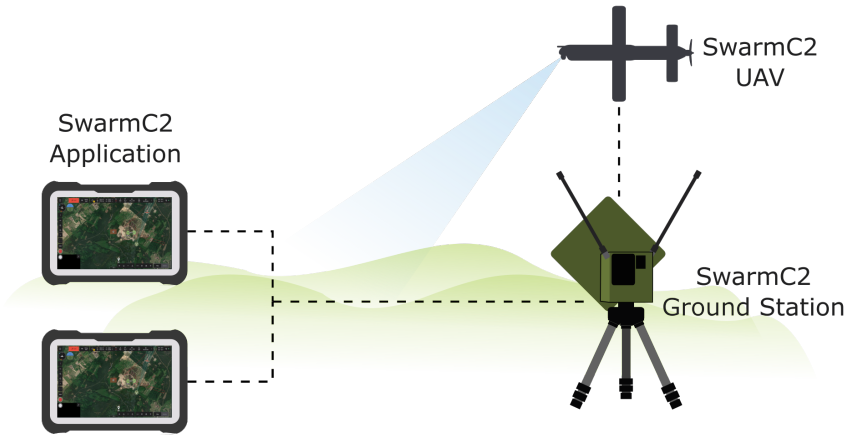


Figure 2.1: SwarmC2 Setup

The application operates in one of two modes depending on whether the operator device has a connection to the ground station:

Operation mode
: the operator device is connected to the ground station. Full application capabilities are available.

Standalone mode
: the ground station is not reachable (ground station not deployed, powered off, or outside network range). A subset of application functions is available locally on the device.

2.1 Access and Installation

SwarmC2 is an application installed on the operator device. It runs in a browser and can be accessed from any device connected to the ground station network. Once installed, it works locally on the device — no active connection is required to launch it.

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NOTE: **SwarmC2** comes pre-installed on the tablets supplied with the ground station. Google Chrome is the recommended browser.

2.2 Standalone Mode

In standalone mode the application works entirely on the device. You can plan missions, set the ground station position, and adjust preferences. Controls that require the ground station are unavailable until you return to operation mode. When the connection is restored, changes made are synced automatically.

2.3 Operation Mode

Operation mode is active when the ground station is connected and you are signed in. It covers two levels of access depending on what is connected:

Ground station configuration
: with only the ground station available, you can configure radio and antenna settings, network connectivity, and external system integrations.

UAV operations
: once a UAV connects to the ground station, live telemetry, commanding, and flight controls become available.

All connected operator devices share the same state, kept in sync through the ground station.

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3 Interface

The GCS tablet comes with **SwarmC2** software pre-installed. No additional setup is required. Simply launch the software to begin.

After launch, the **main window** is displayed. By default, **SwarmC2** opens in **Payload layout**.

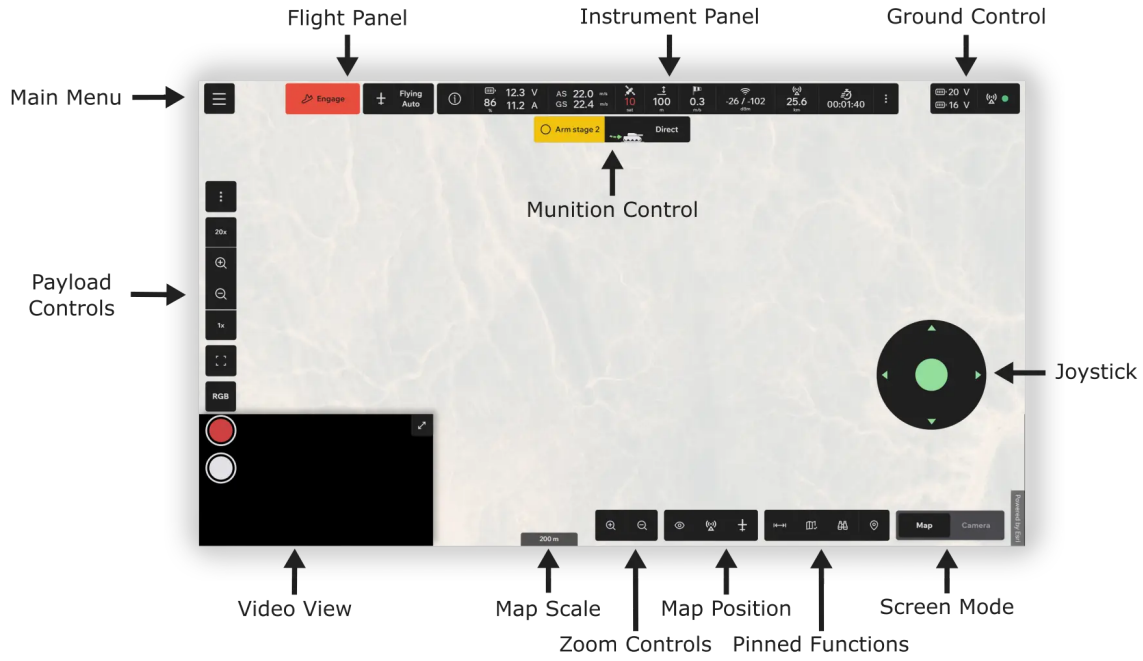


Figure 3.1: Payload layout main window

The **main window** contains the primary controls and information required during operation.

The following interface elements are visible in the **main window**:

1. **Main menu** – Provides access to configuration, navigation, and other **SwarmC2** functions.
2. **Flight panel** – Displays the main flight controls and current UAV state.
3. **Instrument panel** – Displays live telemetry data from the UAV and provides access to related actions.
4. **GCS panel** – Displays ground control station battery information and provides access to GCS-related functions.
5. **Virtual joystick panel** – Touchscreen interface used for payload control or manual UAV control, depending on the selected layout.
6. **Screen mode panel** – Switches the main screen between map view and camera view.
7. **Pinned functions panel** – Displays functions pinned for quick access.
8. **Map position panel** – Provides shortcuts for centering the map on key positions.
9. **Zoom controls panel** – Provides controls for zooming the map in and out.

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10. **Map scale** – Shows the scale of the current map zoom level.
11. **Video/Map panel** – Displays the map or live payload view, depending on the selected screen mode.
12. **Payload control panel** – Displays controls for the current UAV payload.
13. **Munition control panel** – Displays ESAD and engagement state and provides controls for both.

Detailed descriptions of these interface elements are provided in the following sections.

3.1 Main Menu

The **Main menu** provides access to configuration, navigation, and other **SwarmC2** functions.

It includes the following sections:

1. **Display settings** – Used to configure pilot/operator layout and map-related settings.
2. **General settings** – Used to configure general application settings.
3. **Configuration** – Used to configure connectivity and integration settings.
4. **Advanced configuration** – Used for advanced system setup, such as radio binding.
5. **Navigation** – Provides shortcuts to available **SwarmC2** functions that can be pinned to the main window.
6. **Version** – Displays the current **SwarmC2** application version.

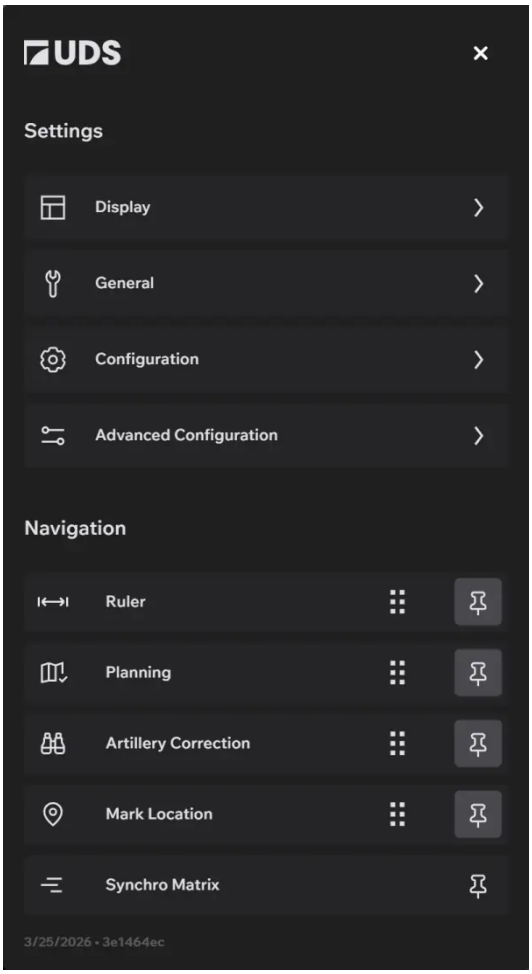


Figure 3.2: Main menu.

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3.1.1 Display Settings

The **Display settings** section is used to configure layout and map display options.

It includes the following options:

1. **Use Offline Maps** – Enables the use of locally stored maps.
2. **Toggle Fullscreen** – Enables full screen mode.
3. **Map View** – Controls which map layers are displayed:
   a. **All** – Displays all available layers.
   b. **Local** – Displays local operational elements such as UAV icons, points of interest, and Five Ws.
   c. **Tasking** – Displays map drawings and overlays related to current mission planning or execution, such as waypoints, paths, and targets.
   d. **Permanent** – Displays permanent or semi-static overlays such as CFF, Guided Loiter zones, and antenna line-of-sight lines.
   e. **Geofence** – Displays all configured geofences that define UAV flight boundaries.
   f. **Zoom controls** – Show or hide map zoom controls, allows the user to zoom map in or out for example using a pen.
4. **Layout** – Selects which predefined layout is used:
   a. **Payload** – Layout optimized for payload controls.
   b. **Pilot** – Layout optimized for aircraft controls.
5. **Advanced** – Enables advanced options.

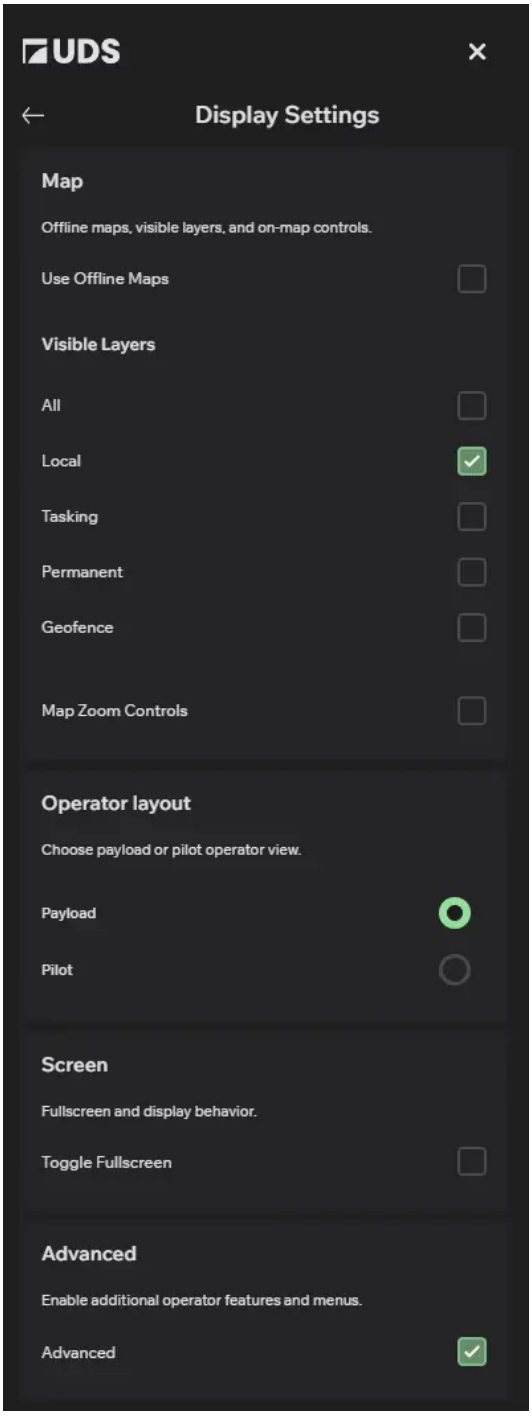


Figure 3.3: Display settings

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3.1.2 General Settings

The **General settings** section is used to configure general application settings and perform system actions.

It includes the following options:

1. **Coordinate format** – Selects how coordinates are displayed in the application.
2. **Map Service** – Checks whether the application can access the map service.
3. **Upload map** – Uploads a map for offline use in MBTiles format.
4. **Clear App Data** – Clears all application data and settings.
5. **Logout** – Logs out the current user from the application.

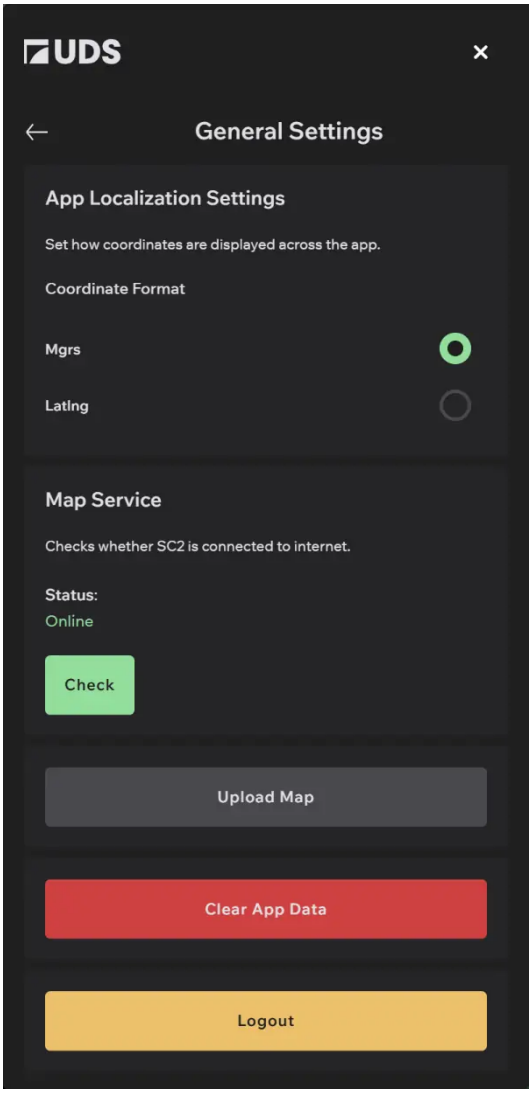


Figure 3.4: General settings

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3.1.3 Configuration

The **Configuration** section is used to configure connectivity and integration settings.

It includes the following options:

1. **Tech support remote access** – Enables or disables secure remote access for technical support sessions.
2. **VPN setup** – Configures and monitors VPN connectivity for secure network access.
3. **Wi-Fi** – Configures the Wi-Fi module for internet access and displays connection status.
4. **TAK credentials** – Configures TAK integration settings, credentials, and connection status.
5. **Video Stream Setup** – Configures video stream connection settings and output behavior.
6. **Training mode** – Enables or disables the GPS plane icon for training/simulation purposes. The icon is hidden while Training Mode is active.

3.1.4 Advanced Configuration

The **Advanced configuration** section is used for advanced system setup tasks.

It includes the following options:

1. **Bind radios** – Links the UAV radio with the GCS radio to enable secure communication between radios.

3.2 Main Window Panels

The following panels are visible in the **main window**.

3.2.1 Flight Panel

The **Flight panel** displays the main flight controls and current UAV state.



Figure 3.5: Flight panel

It includes the following elements:

Flight Command
Used to arm or engage the UAV, depending on the current state.

Flight Mode
Displays the current flight mode and allows selecting another mode from the dropdown list.

The list of available flight modes depends on system configuration and UAV state.

3.2.2 Instrument Panel

The **Instrument panel** displays live telemetry data from the UAV and provides access to related actions.

The three-dot button at the end of the panel opens the **UAV side panel**.

Below the **Instrument panel**, an additional bar shows **Munition Control** and **pinned** telemetry fields selected in the telemetry view.

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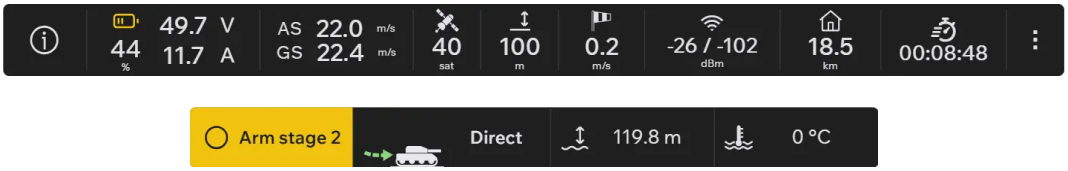
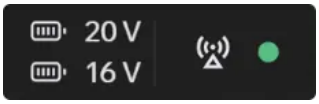


Figure 3.6: Instrument Panel

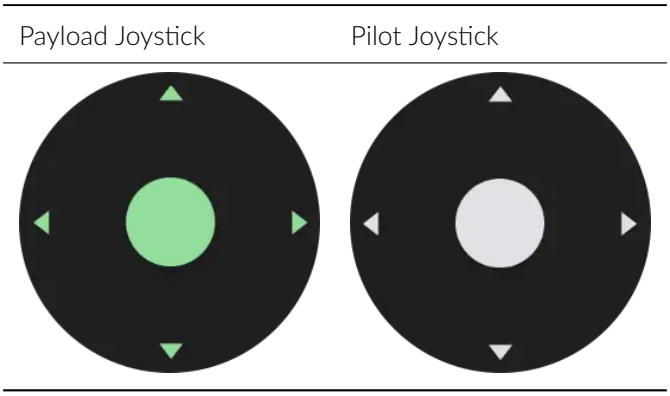
3.2.3 GCS Panel

The **GCS panel** displays ground control station battery information and provides access to GCS-related functions.



Selecting this panel opens the **GCS view**.

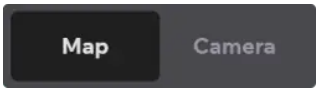
3.2.4 Virtual Joystick Panel



The **Virtual joystick panel** is a touchscreen interface used for payload control or manual UAV control, depending on the selected layout.

3.2.5 Screen Mode Panel

The **Screen mode panel** is used to switch the main screen between map view and camera view.



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3.2.6 Pinned Functions Panel

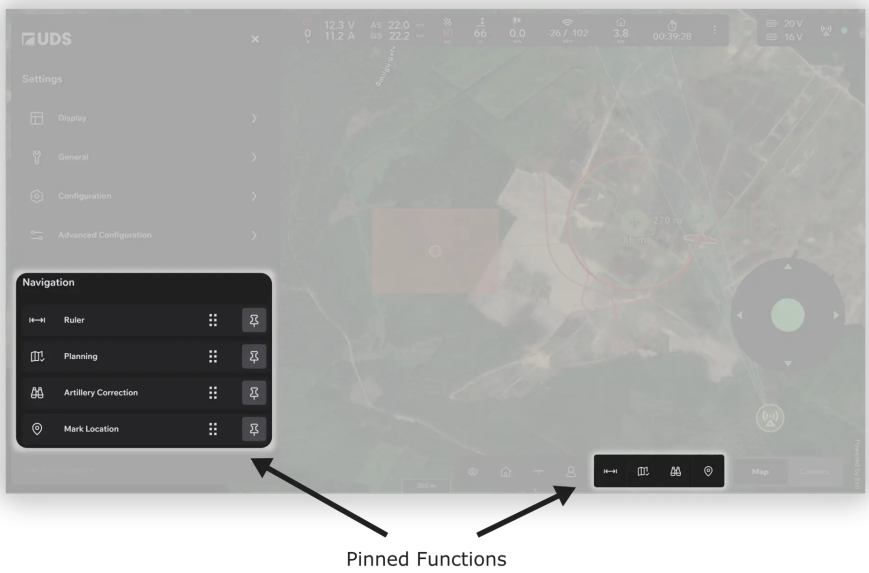


Figure 3.7: Pinned functions

The **Pinned functions panel** displays shortcuts to selected **SwarmC2** functions pinned from the **Main menu**.

Pinned functions appear in the main window for quicker access during operation.

The following functions can be pinned to this panel:

Ruler – Measures distance between points.

Planning – Opens the mission planning menu.

Artillery Correction – Opens the interface for artillery correction using payload live video feed.

Mark Location – Marks a location on the map using input coordinates.

3.2.7 Map Position Panel

The **Map position panel** provides shortcuts for centering the map on key positions.

It includes the following controls:

Camera View – Centers the map on the camera view point.

Ground Station – Centers the map on the ground station position.

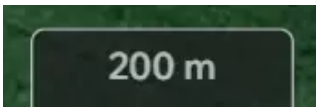
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Plane – Centers the map on the plane position.

My Location – Centers the map on the operator's current position.

3.2.8 Map Scale

The **Map scale** shows the scale of the current map zoom level.



3.2.9 Video/Map Panel

The **Video/Map panel** displays the map or live payload view, depending on the selected screen mode.

The video window can be resized by dragging its top-right corner.

3.2.10 Payload Control Panel

The **Payload control panel** displays controls for the current UAV payload.

3.3 Main Window Layouts

The **SwarmC2** main window can be configured in two layouts: **Payload layout** and **Pilot layout**.

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3.3.1 Payload Layout



Figure 3.8: Payload layout main window

The **Payload layout** is optimized for payload operation and observation tasks.

In this layout, the **Virtual joystick panel** is used for payload control.

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3.3.2 Pilot Layout

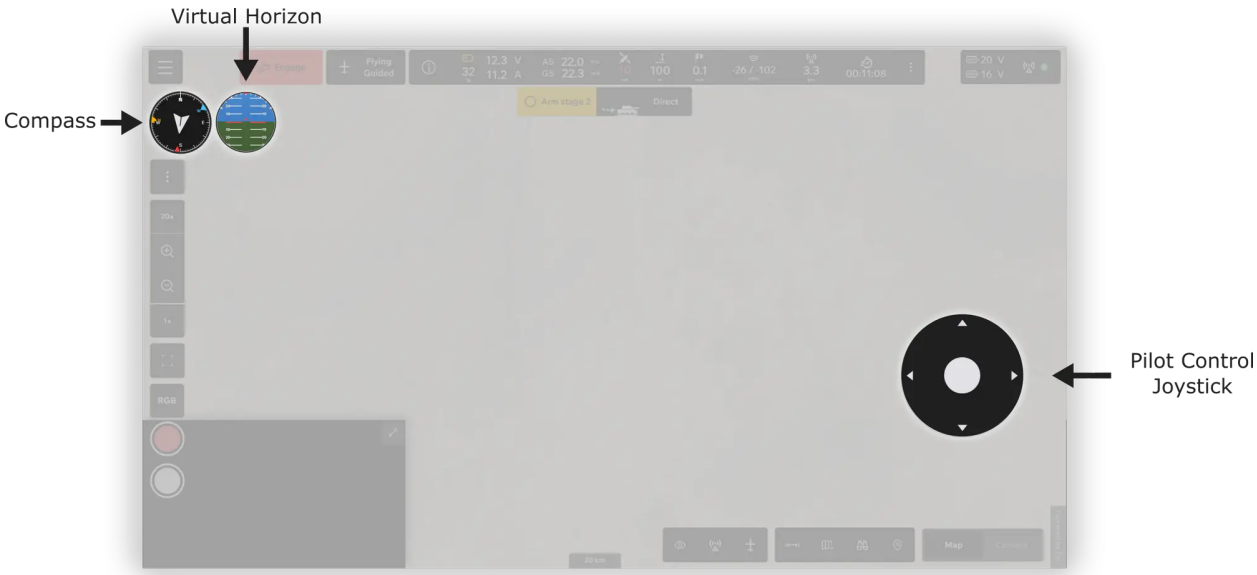


Figure 3.9: Pilot layout main window

The **Pilot layout** is optimized for piloting and flight control.

In this layout:

Virtual Horizon/Compass panel
Displays the current UAV heading and virtual horizon.

Virtual joystick panel
In **Cruise** mode, the joystick is shown in white and is used to control vehicle heading and altitude.

3.4 Map Controls

Main map view has multiple functions for safe and efficient operation of UDS UAVs. This includes direct vehicle and payload controls, as well as multiple map-related functions and markers.

Map can be manipulated with both touch and mouse controls:

Pan the map
Press and drag a finger or mouse across the screen.

Zoom the map
Pinch the screen using two fingers, use the scroll wheel while hovering over the desired area, or use the + and − buttons in the zoom panel.

Open function menu
Tap over a desired area with mouse or finger.

Zoom Controls
Use + and - buttons when **Zoom controls** are enabled through App Menu.

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3.4.1 Symbols

The following symbols appear on the map to indicate key objects during operation.



Plane – marks the current UAV position on the map.



Home position – marks the UAV home / RTL location on the map. The user is allowed to change it during flight.



Marker – marks the point selected on the map.



POI – marks the location of a Point of Interest (POI).



Antenna – marks the antenna location on the map.

3.4.2 Map View Layers

Different map layers can be configured to be displayed on the main map view. To configure them:

1. open **main menu** panel.
2. go to **display settings.**
3. press on checkmark boxes for which layers you want to be shown.

3.4.3 Function Menu

When pressing anywhere on the map, a marker will be placed and a function menu will appear displaying all available functions which require coordinates to function. Main functions include:



Center Map – centers map to clicked coordinates



Add Five W – adds a 5W report to the map.



Add POI – adds point of interest marker on map



Call For Fire – generates a CFF request on selected coordinates and copies formatted string to clipboard



Set Home Position – sets the selected point as the UAV's default **RTL** destination while the UAV is flying.



Observe – points the camera payload towards the selected coordinates and maintains continuous track.

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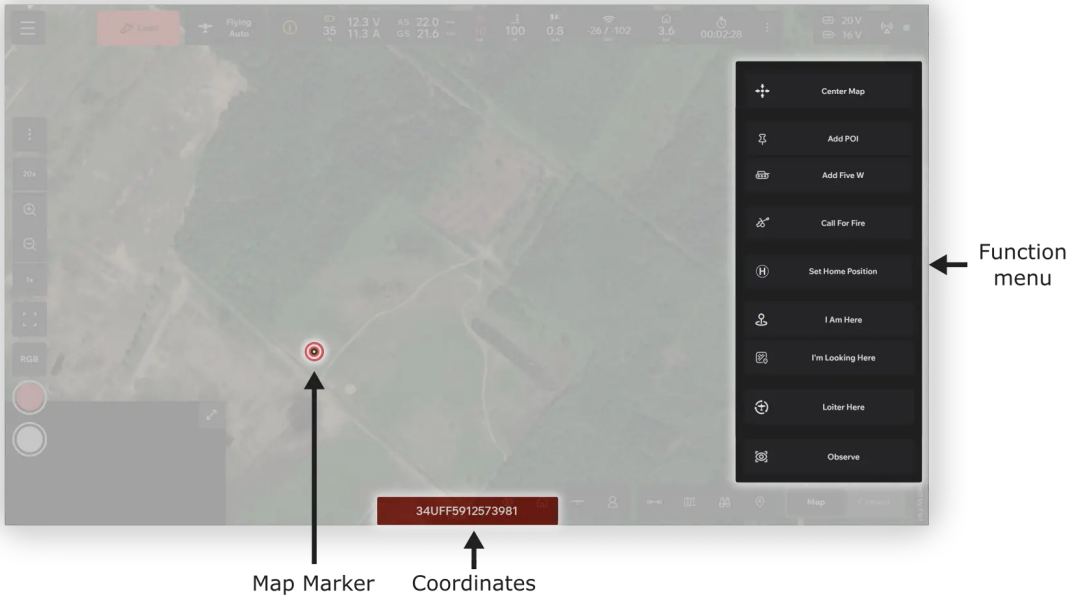


Figure 3.10: Function menu

Loiter Here – set new loiter in this position.

I'm Here – sets the UAV's location to this point when GNSS is unavailable.

I'm Looking Here – When clicked, it sends the current coordinates to the UAV, prompting it to recalculate its position in GNSS-disabled environments based on camera zoom and other factors

Edit POI/5W – Ability to edit / delete POI or 5W

NOTE: Coordinates are in a format specified from the general settings tab.

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4 Authentication

This chapter describes how operators sign in to **SwarmC2** when using the payload operator application with the ground control station, how sessions end, and how the mandatory **first-time password change** works when accounts still use the factory default password.

Green circle in top right corner indicates whether Ground Station is connected.

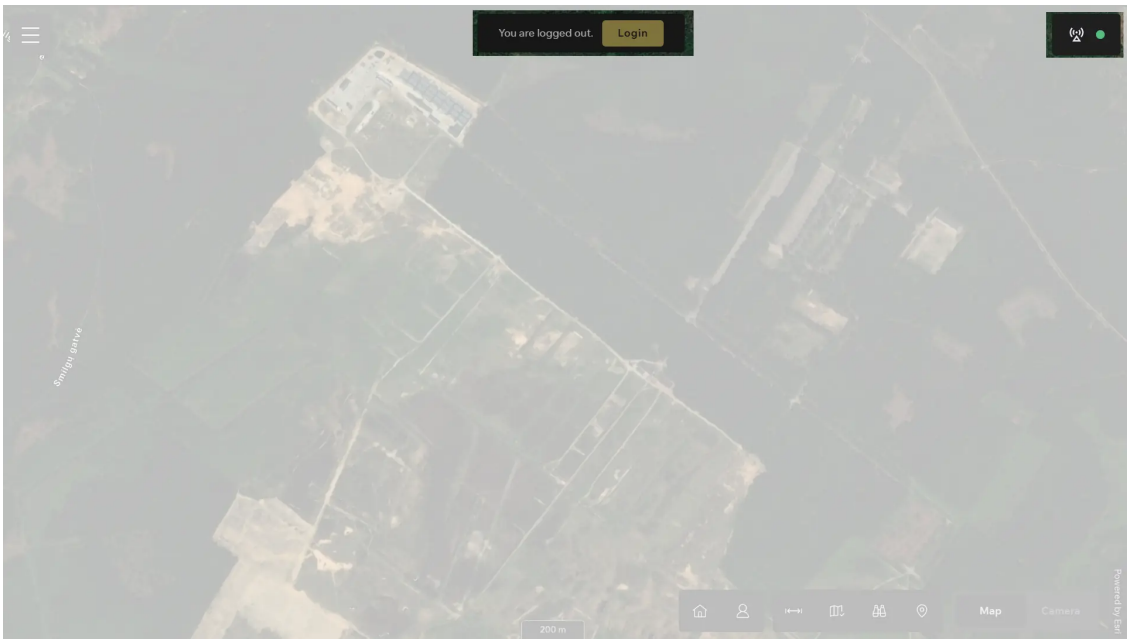


Figure 4.1: Logged Out Overview

4.1 When sign-in is required

After **SwarmC2** has launched and the tablet has connected to the **ground station**, the main map and interface are available, but server-backed user data and preferences are only loaded after a successful sign-in.

If the ground station link is up and you are **not** signed in, a notice appears top of the page with a **Login** control. Use that control to open the sign-in dialog.

NOTE: If the ground station is not connected, this sign-in notice does not appear. Establish the radio link first, then sign in.

4.2 Signing in

The application distinguishes two built-in operator identities, identified by **call sign**:

- **Payload** – Payload operator layout and functions.
- **Pilot** – Pilot operator layout and functions.

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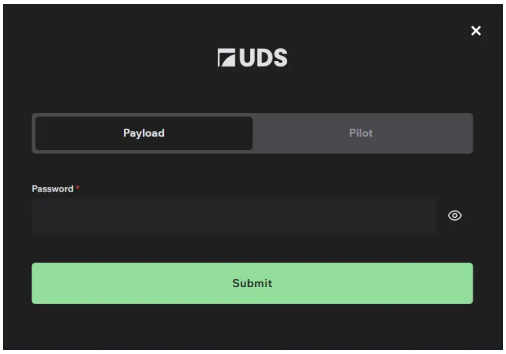


Figure 4.2: Login Pop-up window

NOTE: These are not separate accounts you create in the tablet; they are fixed roles the server recognises. Your organisation sets the password for each call sign on the ground-station service.

4.2.1 Login dialog

1. Use the horizontal toggle to select **payload** or **pilot** according to the role you are using for this session.
2. Enter the password for that call sign.
3. Choose **Submit** to authenticate.



Use the Eye button to hide or show the password.



You may close the dialog with the **close** control in the corner of the sign-in window. That returns you to the map without signing in; server-backed personal settings remain unavailable until you sign in successfully.

4.2.2 After a successful login

On success, the application stores authentication in the browser session, loads your user profile and preferences, and returns you to the normal **main window** workflow.

4.2.3 Sign-in errors

The application surfaces two common problems:

- **Invalid username or password** – The call sign or password does not match what the server expects. Check caps lock, the selected role (**payload** / **pilot**), and the password provided by your administrator.
- **Ground station unavailable** – The client could not reach the authentication service (for example, network outage). Resolve connectivity, then try again.

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4.3 Changing the factory default password

For a new or freshly initialised system, each built-in call sign may still use the **factory default password** configured on the ground-station service (deployment-specific; not documented here). **SwarmC2** does not allow continued use of that default once it recognises it: the first successful check of the default password does **not** complete a normal login.
Instead, the application switches to the **set new password** screen. You must choose a strong password before you can use the software in the usual way.

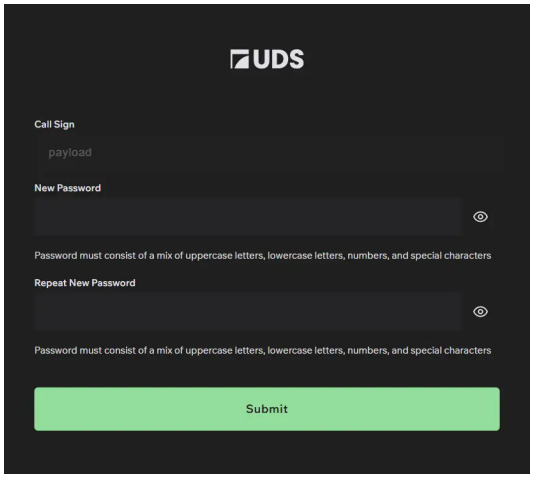


Figure 4.3: Password Reset

IMPORTANT: Complete this step promptly. The server only accepts the password-change request for a short time after it prompted you (on the order of a few minutes). If that window expires, sign in again with the default password to obtain a new change-password opportunity.

4.3.1 Set new password form

The form shows the **call sign** for the role you just authenticated; it is fixed for this step so it matches the server's expectation.

Enter the **new password** twice. The password must satisfy the strength rules used by the application, including at minimum:

- At least **8** characters.
- At least one **uppercase** letter.
- At least one **lowercase** letter.
- At least one **digit**.
- At least one **special character** from the supported set (for example `!@#$%^&*()_,.?":{}|<>`).

Submit the form when the fields are valid. On success, you are returned to the **login** dialog; sign in again with the **new** password to start your session.

IMPORTANT: If the server rejects the change (for example, the short-lived authorisation from the previous step has expired), you will be sent back to login with an appropriate message. **Use the default password** once more if you still need to perform the initial change, or **use your new password if the change already succeeded**.

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4.4 Signing out

To end the session and clear authentication from the tablet, use **Main menu** → **General settings** → **Logout** (see the **Interface** chapter). Logging out clears local session data associated with the application on that device; sign in again when you need server-backed settings.

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5 UAV Status and Settings

UAV Panel provides connected UAV detailed telemetry and log information as well as parameter and sensors control. It is accessed in two ways:

1. by clicking on three dots **in Instrument panel** on the top of the screen
2. by clicking on three dots on the target UAV in UAV panel

The UAV Panel header contains:

1. **ID**. Vehicle Identification tag. It is saved locally for each UAV.
2. **UAV Type**. Currently selected UAV type.
3. **Pencil.** Button for changing identification tag name.
4. **Cross**. Panel close button

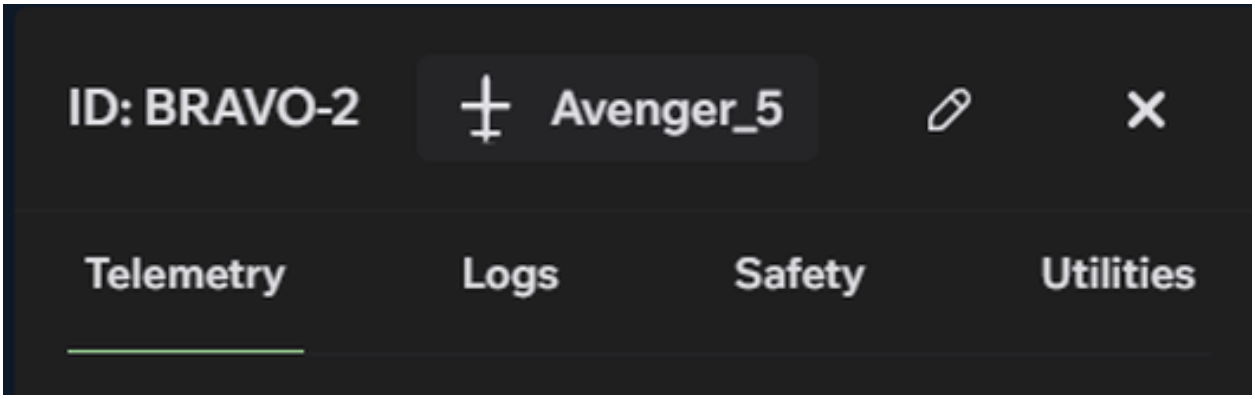


Figure 5.1: UAV Panel header

To rename the UAV tag press the pencil button and new tag input will appear. By pressing the **check** button new provided identifier will be saved locally.

UAV panel provides four main tabs:

1. **Telemetry**. Provides all available UAV telemetry data.
2. **Logs**. A list of all system messages with timestamp and severity level.
3. **Safety**. Safety parameters configuration.
4. **Utilities**. Sensor calibration and additional UAV actions.

5.1 Telemetry

Telemetry panel provides drone current telemetry data. Each parameter is provided with icon and value and can be pinned on the main window using the **pin** button.

Altitude MSL. Drone altitude above Mean Sea Level (MSL), in meters.

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Air Speed. Speed of the drone is relative to the surrounding air, in meters per second.



Ground Speed. Horizontal speed of the drone over the ground, in meters per second.



Volts. Current total pack battery voltage.



Amps. Instantaneous current draw from the battery, in amperes (A).



Wind. Estimated wind speed (in meters per second) and direction affecting the drone.



GPS Count. Number of GNSS satellites currently being tracked.



Distance to return. Horizontal distance from the drone to the defined return point, in kilometers.



ESC temperature. Temperature of the Electronic Speed Controller, in degrees Celsius.



Motor temperature. Temperature of the motor, in degrees Celsius.



Signal strength. Quality of the control/telemetry link between the drone and ground station, shown as RSSI **signal** and **noise** levels.



Flight time. Duration of the current flight since take off.

5.2 Logs

A list of all system messages with timestamp and severity level. Messages are shown since the selected UAV was connected to **SwarmC2**.

There are three levels of severity:

Info
: Normal system messages that describe regular operations or state changes. No action is required; these are for monitoring and traceability.

Warning
: Indicates a potential issue or degraded condition that may require attention but does not immediately affect flight safety or system operation.

Critical
: Serious fault or unsafe condition that can affect flight safety or core system functions. Requires immediate attention and may need emergency action.

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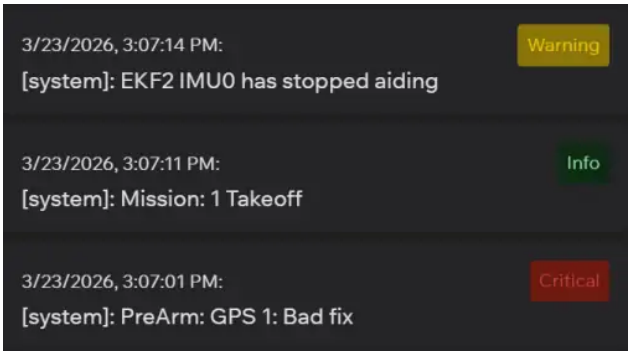


Figure 5.2: System log messages

5.3 Safety

Safety defines how the aircraft behaves in failsafe and emergency situations (RTL behavior, link/GPS failsafes, mission handling) to keep operations predictable and safe. When the Safety Parameters panel is opened, current UAV parameters are loaded. After making changes, press **Save** to write the updated parameters back to the UAV.

5.3.1 Return To Launch (RTL)

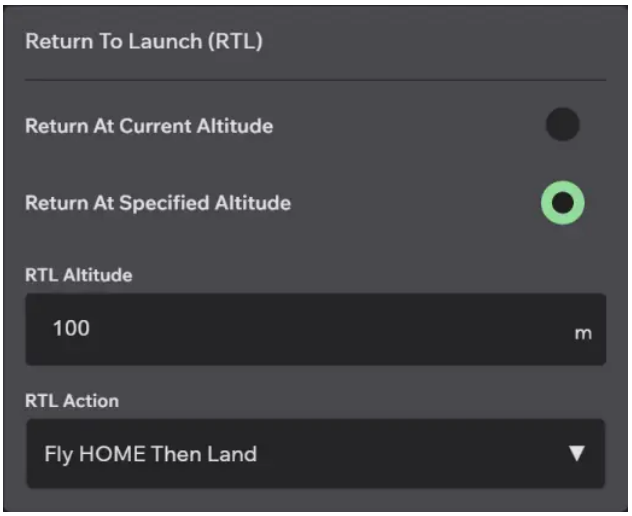


Figure 5.3: RTL configuration

Return at current altitude
When RTL is triggered, the drone flies back home at its current altitude and then follows the configured landing sequence.

Return at specified altitude
When RTL is triggered, the drone first climbs/descends to the RTL Altitude and then flies back to home at that altitude.

RTL Altitude
Preset altitude the drone will use during RTL when **Specified Altitude** is selected.

RTL Action

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1. **Disable** – RTL will not be triggered.
2. **Fly HOME then land.** – On RTL, the drone flies to home point or rally point and loiters unlimited.
3. **Go directly to landing sequence.** On RTL, the drone skips flying to home and immediately enters the configured landing sequence.
4. **OnlyForGoAround**. On RTL, starts a go-around procedure.

Rally points planned in the mission may be used as alternate RTL destinations when they are nearer than the home or return point and within configured range limits. See *Rally Points*.

5.3.2 Failsafe Triggers

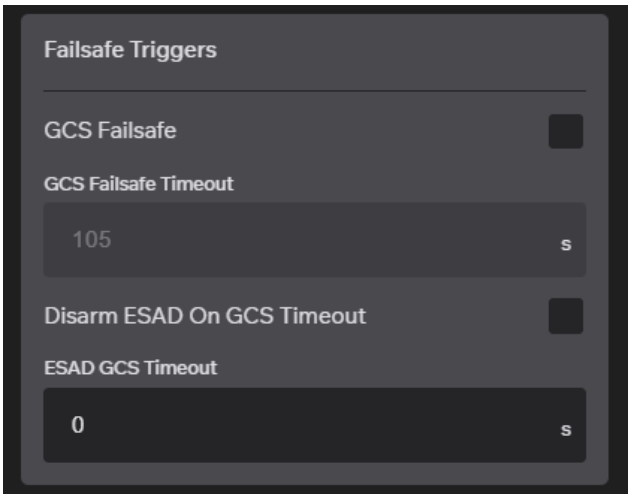


Figure 5.4: Failsafe action triggers

GCS Failsafe
Enables or disables failsafe actions when the connection to the Ground Control Station is lost

GCS Failsafe timeout
Time period (in seconds) the GCS link must be lost before the GCS failsafe is triggered.

Disarm ESAD on GCS timeout
When enabled, disarms ESAD if no GCS ping is received for the configured timeout while the vehicle is armed.

ESAD GCS timeout
Time period (in seconds) without a GCS ping before ESAD is disarmed. Set to **0** to disable.

5.3.3 GPS

Automatic drift calculation
Enables automatic estimation of GPS position drift to improve position-hold and navigation accuracy.

GPS Failsafe
Enables or disables failsafe actions when GPS quality is lost or becomes unreliable.

GPS Failsafe timeout
Time period (in seconds) that GPS must be in a failed/poor state before the GPS failsafe is triggered.

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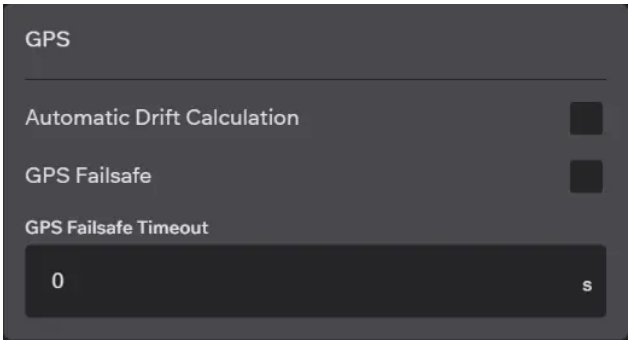


Figure 5.5: GPS safety configuration

5.3.4 Mission Options

Delete mission on reboot

When enabled, the stored mission is cleared automatically each time the flight controller reboots.

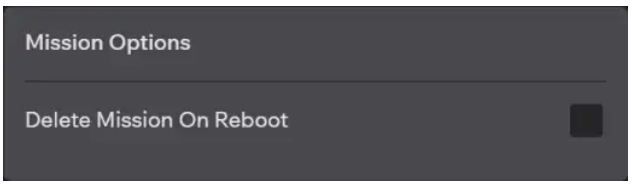


Figure 5.6: Delete mission on reboot

5.3.5 Logging

Log Enable

Turns onboard flight data logging on or off

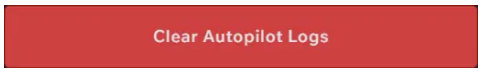


Figure 5.7: Log enable

5.3.6 Safety Actions

Clear Autopilot Logs

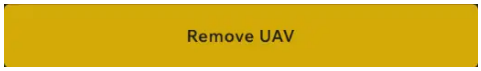
Deletes all stored autopilot log data from the flight controller. Use to free space or remove old logs; this action cannot be undone.



Remove UAV

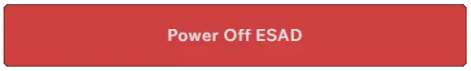
Drops this aircraft from the ground station's active list until it shows up again with new telemetry. Use it when aircraft state is stale – and you want to reset it.

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Power off ESAD

Cut the power supply off for ESAD.



NOTE: If this button is pressed while the plane is airborne, there is no way to recover the ESAD.

5.4 Utilities

5.4.1 Compass Calibration

When flying for the first time in a new location, vehicle may return a **compass inconsistent** error. To fix this, simply calibrate compass using built-in SwarmC2 functionality



Figure 5.8: Compass calibration menu

To calibrate compass:

1. Select vehicle that needs to be calibrated.
2. Go to **Utilities** menu.
3. Pick up vehicle and move to a **clear area**.

CAUTION: Make sure there are no large magnetic objects near the UAV when doing compass calibration, like buildings, cars, power poles, as calibration may fail.

NOTE: It is recommended to remove any electronic devices from the person that is doing the calibration.

4. Start compass calibration.
5. Rotate the vehicle through all its axes. Rotate the vehicle steadily, making 1-3 turns throughout each axis.
6. Repeat rotations on a random axis until the progress bar is complete.
7. Reboot the vehicle.

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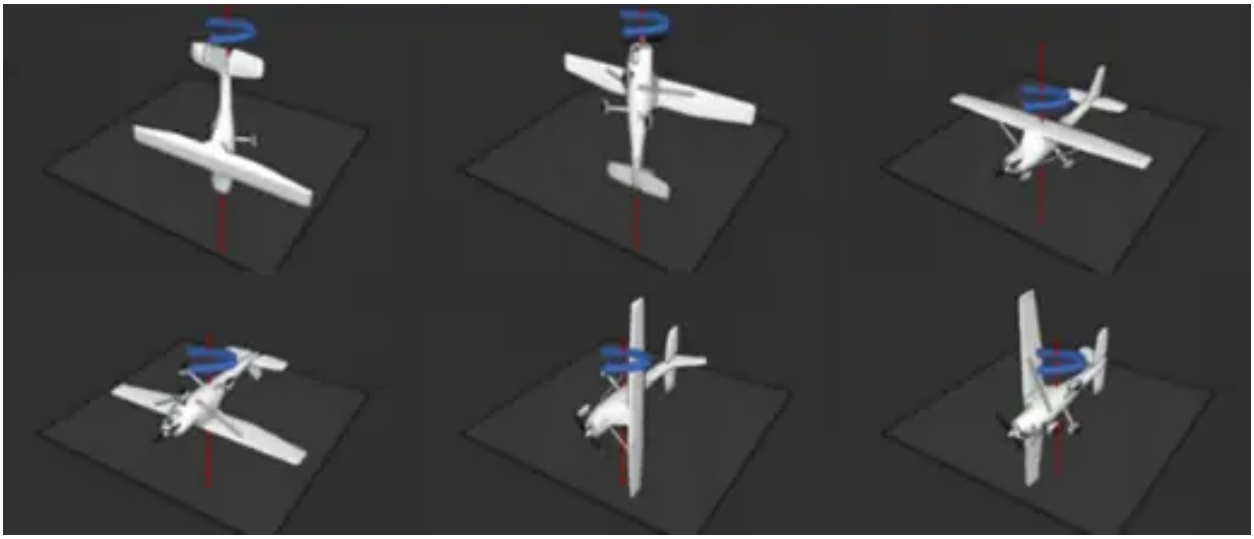


Figure 5.9: Compass calibration rotation axes

8. Repeat calibration until both compasses are calibrated in the **green zone.**

CAUTION: If an error occurs during calibration or the bar stops moving, reboot the vehicle and try again.

5.4.2 Simple Compass Calibration

This is a simplified compass calibration procedure. It is recommended to use the full compass calibration procedure for best results. This compass calibration should only be used if the full compass calibration procedure is not available or cannot be performed.

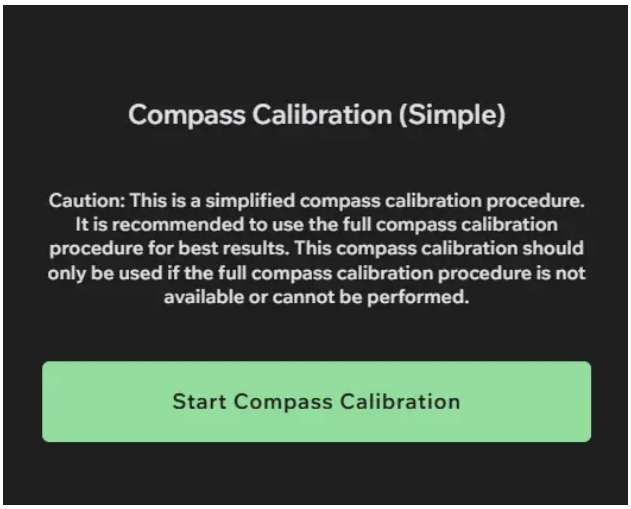


Figure 5.10: Compass calibration menu

To calibrate compass:

1. Select vehicle that needs to be calibrated.
2. Go to **Utilities** menu.

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3. Place the vehicle on a steady surface.
4. Click **Start Compass Calibration**

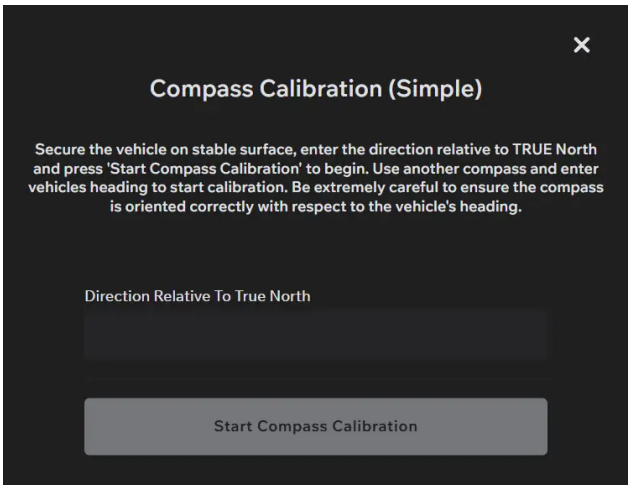


Figure 5.11: Compass calibration menu

5. Use another compass to get exact vehicle's heading (**Relative to TRUE NORTH**)
6. Enter the direction and press Start Compass Calibration to begin

CAUTION: Make sure there are no large magnetic objects near the UAV when doing compass calibration, like buildings, cars, power poles, as compass may produce an incorrect value.

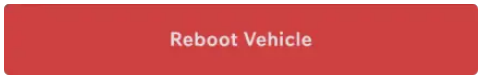
CAUTION: Be extremely careful to ensure the compass is oriented correctly with respect to the vehicle's heading.

5.4.3 Reboot Commands

Reboot commands are used after certain actions that specifically request the UAV or Gimbal to be rebooted.

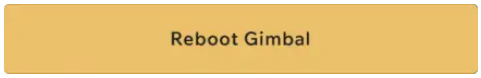
Reboot Vehicle

Autopilot reboot command is used after certain actions that specifically request the UAV to be rebooted, like configuring parameters that change functionality, after finishing sensor calibration, or some fault is detected on startup. Reboot command is never used during flight and this function is always disabled after the vehicle is armed.



Reboot Gimbal

This command power-cycles the payload module. It is used after configuring payload functions that specifically request for it to be rebooted, or can be used during flight if a fault is detected.



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6 Mission Planning

SwarmC2 software supports mission planning functionality to automate UAV control during operations. This includes automatic takeoff and landing sequences, mission waypoints, and safety-related functionality such as geofences, rally points, and return points.

Planning functionality can be opened from the **main menu** or pinned to the **main window**.

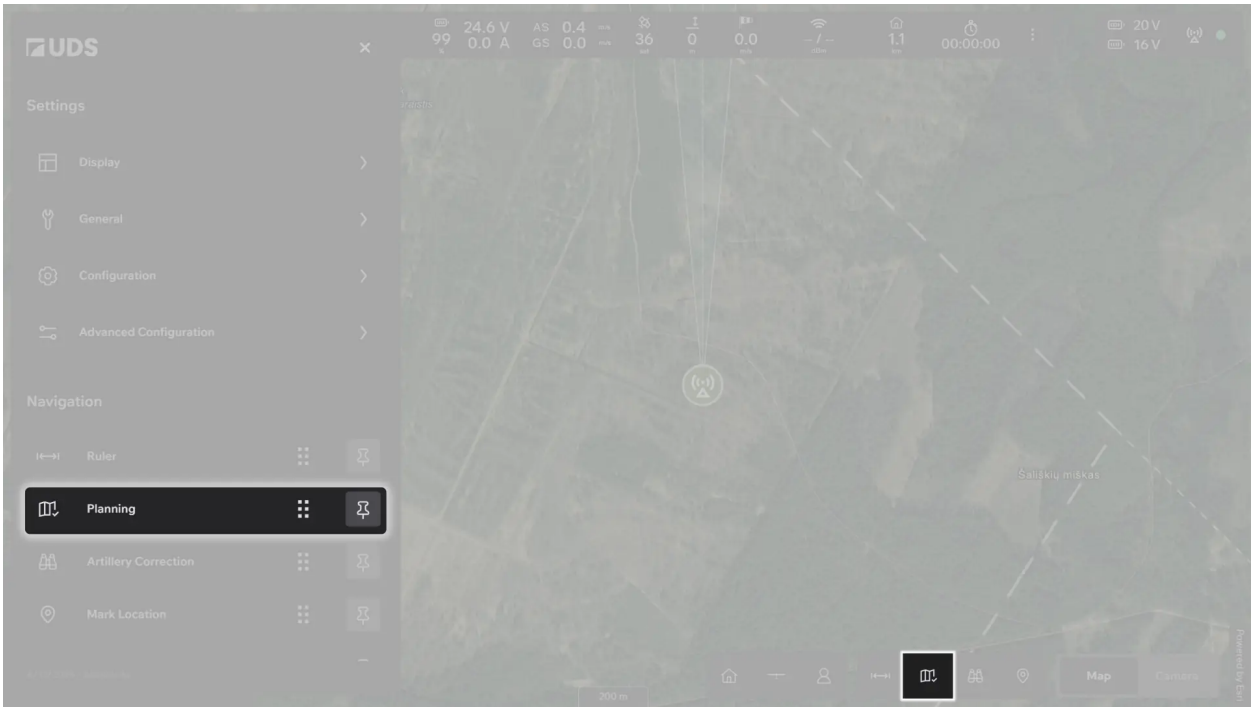


Figure 6.1: Open mission planning

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6.1 Start Screen

When the mission planner opens, the map is centered on the current UAV location, if one is available. Leaving the planner returns to the default map view and refreshes the mission currently stored on the connected UAV.

The start screen provides the main planning actions:

1. ***New Mission*** creates a new empty mission draft.
2. ***Edit Mission*** opens the currently assigned mission for editing. If the UAV has no mission uploaded, this **button** is hidden.
3. ***Saved Missions*** opens missions stored on the current device. A mission can be saved for later use and not yet uploaded to UAV.
4. ***Geofences*** opens the geofence planning view.
5. ***Sync Mission Plan*** fetches the mission currently stored on the connected UAV.

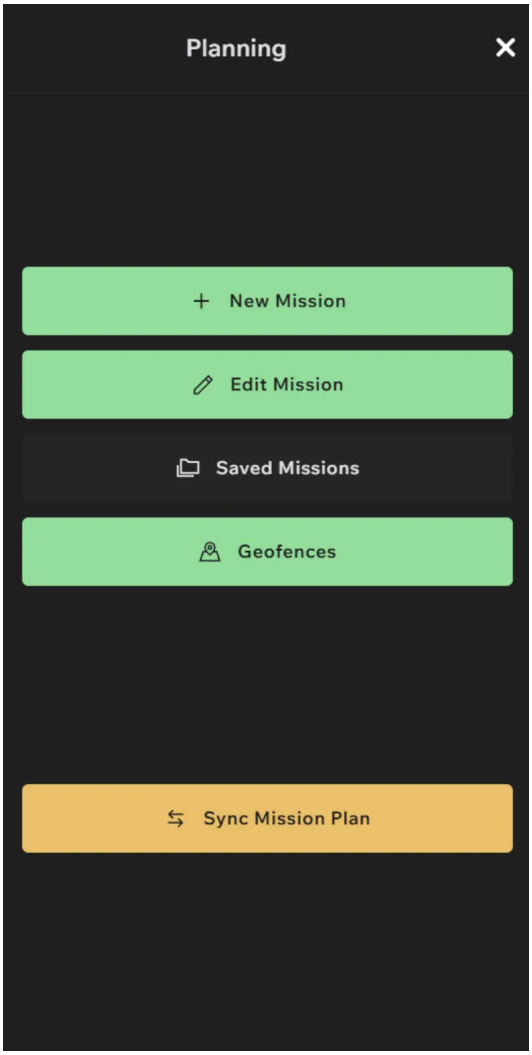


Figure 6.2: Mission planning menu

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WARNING: When a UAV connects to the ground station, **SwarmC2** fetches its configuration. If mission-planning configuration is not available for the supported UAV types, mission-planning actions are disabled.

6.2 Sync Mission Plan

When the UAV first connects to the GCS and is detected by SwarmC2, the software **automatically syncs any pre-existing mission plan** from the UAV, if available. Rally points stored on the UAV are included in this sync.

If changes are made to the UAV mission plan externally or through a second tablet, the system will automatically re-sync the mission plan, but the mission can also be manually synced at any time using **Sync Mission Plan**.

To sync a mission manually:

1. Open the mission planner.
2. Press **Sync Mission Plan**.

6.3 Mission Workflow

The mission planner follows a step-based workflow:

1. Start screen.
2. Vehicle type selection.
3. Mission path editing.
4. Mission upload.

6.3.1 Vehicle Type Selection

When entering the planning window, the software automatically checks which type of UAV is being configured and selects the appropriate mission planning presets.

Currently, SwarmC2 supports:

1. Forecaster ISTAR UAV platform.
2. Avenger 5 Loitering Munition.

When creating a new mission, the user must choose a UAV type before planning can continue. If only one UAV type is available in the current configuration, that type is selected automatically and the planner proceeds directly to the mission path step.

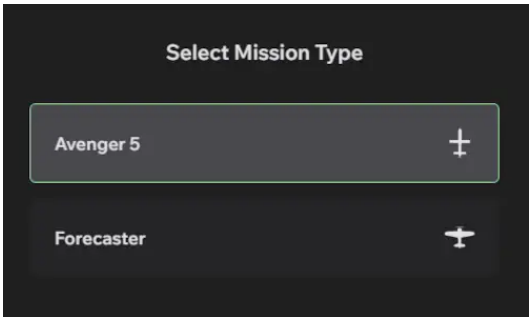


Figure 6.3: Mission type

If a mission plan is being preconfigured for a different type of UAV, or if nothing is currently connected to SwarmC2 software, the vehicle type can be configured as follows:

1. Open the mission planner.
2. Press **New Mission**.
3. Select the mission type from the available options.

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6.3.2 Mission Plan Structure

A mission plan consists of three main flight stages:

1. Takeoff.
2. Waypoints.
3. Return Point.

Each UAV must have a valid mission plan uploaded before flight. A saveable plan must include a takeoff, at least one mission waypoint, and a return point.

Takeoff is a fully automatic sequence in which the UAV climbs to a specified altitude and maintains the heading set during arming. All mission plans must contain a valid takeoff waypoint at the beginning of the mission.

To set a takeoff waypoint:

1. Select the **Takeoff** item in the planning menu.
2. Press on the map where you want the takeoff to begin.
3. Configure the takeoff parameters as appropriate.

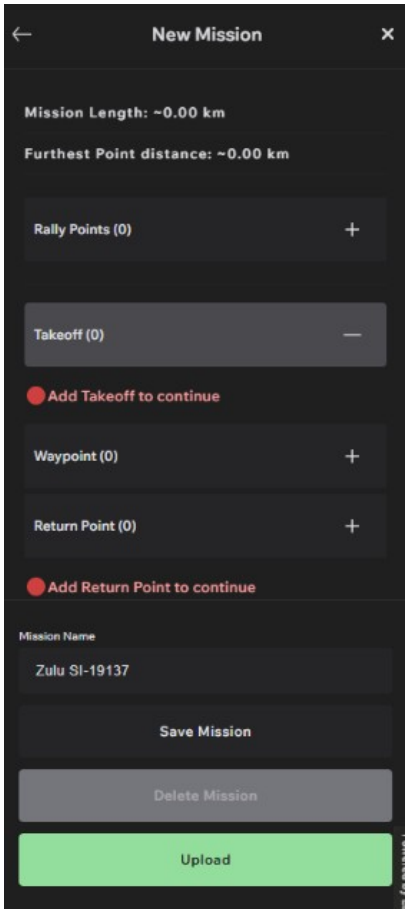


Figure 6.4: Mission planning

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6.3.3 Mission Path Editing

The mission path screen is divided into four expandable groups:

1. **Rally Points**
2. **Takeoff**
3. **Waypoints**
4. **Return Point**

The planner also shows:

1. Total mission length.
2. Furthest point distance.
3. A waypoint altitude summary on the map when the mission contains items.

Users can activate one of the groups and then click on the map to place that type of mission item.

Mission map interaction works as follows:

1. Clicking an existing mission item activates it for editing.
2. Clicking a midpoint between existing items inserts a new regular waypoint at that position.
3. Clicking on the map while a group is active creates a new mission item of that type.
4. Only one takeoff item is allowed. If a takeoff already exists, the next takeoff placement becomes a regular waypoint.
5. Only one return point is allowed. If a return point already exists, the next placement will move it to the new location.

After a takeoff point is placed in a new mission, the planner automatically switches the active placement mode to regular waypoints.

When an existing mission item is selected, its parameter editor opens in the planner panel. The available parameters depend on the selected item type.

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6.3.4 Rally Points

Rally points are optional alternate safe locations that can be uploaded with a mission plan. They are stored separately from mission waypoints and are **not flown during normal AUTO mission execution**. Instead, the autopilot may use them during **RTL** or other failsafe recovery when a rally point is nearer than the home or return point and within configured range limits.

Use rally points when the default home or return location is unsuitable for recovery, but a nearer clear area — such as an alternate field or open ground behind terrain — provides a safer hold or landing option.

NOTE: Rally points are optional. A mission can be saved and uploaded without any rally points.



Figure 6.5: Rally points on the mission map

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Adding Rally Points

To add rally points to a mission:

1. Open the mission path editor.
2. Expand **Rally Points** and activate the group for placement.
3. Click on the map at each desired rally location. Multiple rally points are allowed.
4. Configure the altitude for each rally point as required.

When a rally point is placed, its default altitude is taken from the most recently placed mission item, or from the UAV default cruise altitude if no prior item is available.

Editing and Removing Rally Points

To change an existing rally point:

1. Click the rally point marker on the map, or select it from the **Rally Points** list.
2. Set **Altitude** in the parameter panel. Rally points do not expose waypoint type, loiter radius, or other mission-waypoint parameters.
3. Drag the marker on the map to reposition the rally point.
4. Delete a rally point using the delete control shown when the item is active.

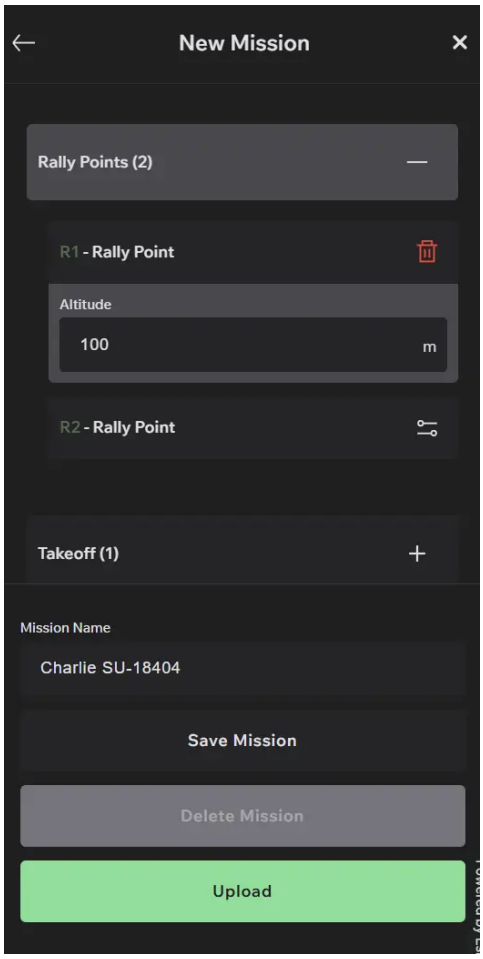


Figure 6.6: Rally Points list in the mission planner

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Map Display

Rally points appear on the map separately from the mission path:

1. Green markers labeled **R1**, **R2**, and so on.
2. A loiter circle around each rally point at the configured rally radius.

The default rally loiter radius for **Avenger** missions is **300 m**.

When a UAV is connected, the displayed circle radius follows the vehicle **RTL loiter radius** loaded from the aircraft.

Rally points are excluded from the mission path line, total mission length, and furthest-point distance. When the mission contains items, the altitude summary chart shows rally points as separate green markers from the main mission path.

Upload and Sync

Rally points are saved with the mission on the local device and uploaded to the UAV together with the mission waypoints. If a mission with no rally points is uploaded, any rally points previously stored on the UAV are cleared.

During **Sync Mission Plan**, rally points currently stored on the connected UAV are fetched and displayed in the planner along with the rest of the mission.

Operational Behavior

During **RTL**, the autopilot may divert to the nearest rally point instead of the home or return point when that rally point is closer and within configured range limits. Rally recovery behavior also depends on the **RTL Action** and related settings configured in the UAV **Safety** panel. See *Return To Launch (RTL)* and *RTL Mode* for more information.

6.3.5 Editable Parameters

The mission path editor currently exposes the following parameters:

Altitude
: defines the commanded altitude for the selected mission item. This is available for takeoff, regular waypoint, and rally point items.

Type
: changes a regular waypoint between supported waypoint behaviors:

1. **Waypoint** is a standard transit point.
2. **Loiter unlimited** keeps the UAV circling the point until a later mission action changes it (requires user interaction).
3. **Loiter turns** keeps the UAV circling the point for a defined number of turns.
4. **Loiter to altitude** keeps the UAV circling while climbing or descending to the required altitude.

Exit Loiter from
: controls how the UAV leaves a loiter pattern:

1. **Center** leaves the orbit by passing through the center of the loiter circle.
2. **Tangent** leaves the orbit tangentially from the loiter circle.

Leave Loiter at
: defines how the UAV continues after loitering:

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1. **Direction of next Waypoint** exits the loiter in the direction of the next mission item.
2. **Any Direction** allows exit based on the current direction of travel.

Loiter Radius
sets the size of the loiter orbit in meters.

Turns
sets how many circles are flown for **Loiter turns** waypoint.

Loiter Clockwise
changes the orbit direction for loiter items. Clockwise or Counter-Clockwise

Some mission values are still taken from UAV configuration defaults rather than edited directly in the planner. Examples include default cruise speed, default loiter radius or takeoff pitch.

6.4 Saving and Reusing Missions

6.4.1 Saving Missions

The **mission path** screen allows the user to:

1. Set or edit the mission name.
2. Save the mission to the local device.
3. Delete a stored mission from the local device.
4. Continue to the upload step.

Mission names are limited to 24 characters in the planner.

6.4.2 Saved Missions

Saved missions are grouped by UAV type. Selecting a saved mission loads it into the planner, draws it on the map, and centers the map on the first mission item.

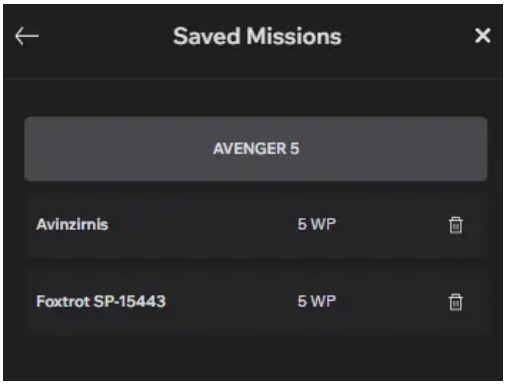


Figure 6.7: Saved missions

6.5 Mission Validation

Before a mission can be saved or uploaded, the planner validates it. A valid mission must have:

1. A selected UAV type.

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2. A mission name.
3. A takeoff item.
4. A return point.
5. At least one mission waypoint.

In addition:

1. All mission items must have coordinates and altitude.
2. Minimum altitude for waypoints is 50m.
3. **Loiter Turns** waypoints must include the number of turns.
4. Loiter waypoint radius must not be lower than the configured default loiter radius for that UAV type.

If validation fails, the planner displays the relevant error messages and the mission cannot be saved or uploaded.

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6.6 Mission Upload

During upload, the planner shows connected UAVs that match the selected mission type.

The currently active UAV is selected automatically if it matches the mission type and is connected. Otherwise, the first connected eligible UAV is selected automatically.

To upload a mission:

1. Open the **Upload** step from the **mission path** screen.
2. Select a connected UAV of the same mission type.
3. Use the confirmation slider to upload the mission.

The upload includes rally points defined in the mission, if any. Each rally point must have valid coordinates and altitude before upload can succeed.

WARNING: If no connected UAV of the correct type is available, the mission cannot be uploaded at that moment. In that case, save the mission in the previous step and upload it later when a matching UAV is connected.

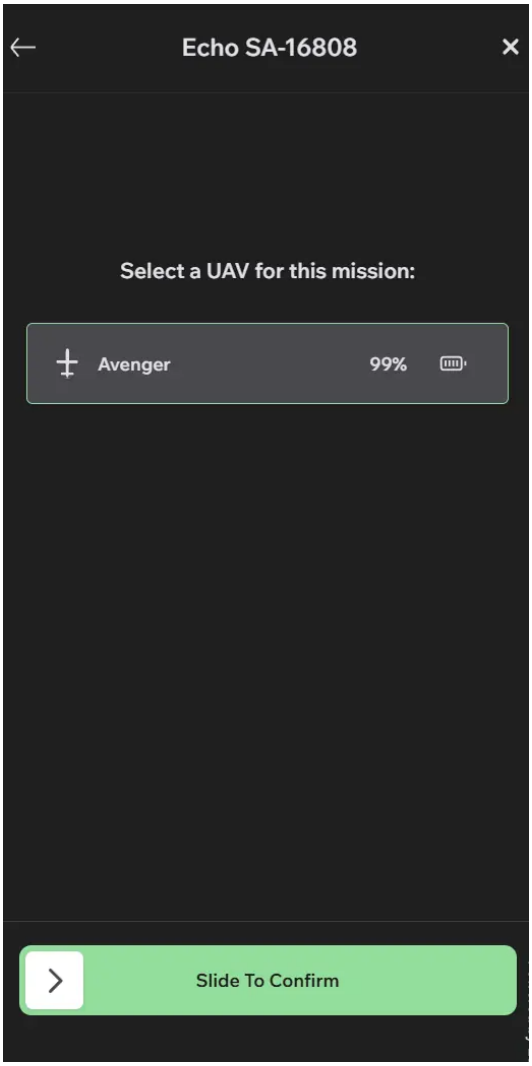


Figure 6.8: Upload mission

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6.7 Geofence Planning

Geofence planning is opened from the mission planner **start screen** by selecting **Geofences**.

The current geofence workflow supports:

1. One or more **polygon geofence groups**.
2. Per-group **inclusion** or **exclusion** behavior.

Geofence edits are managed separately from the normal mission waypoint workflow. The current planner UI exposes polygon geofences.

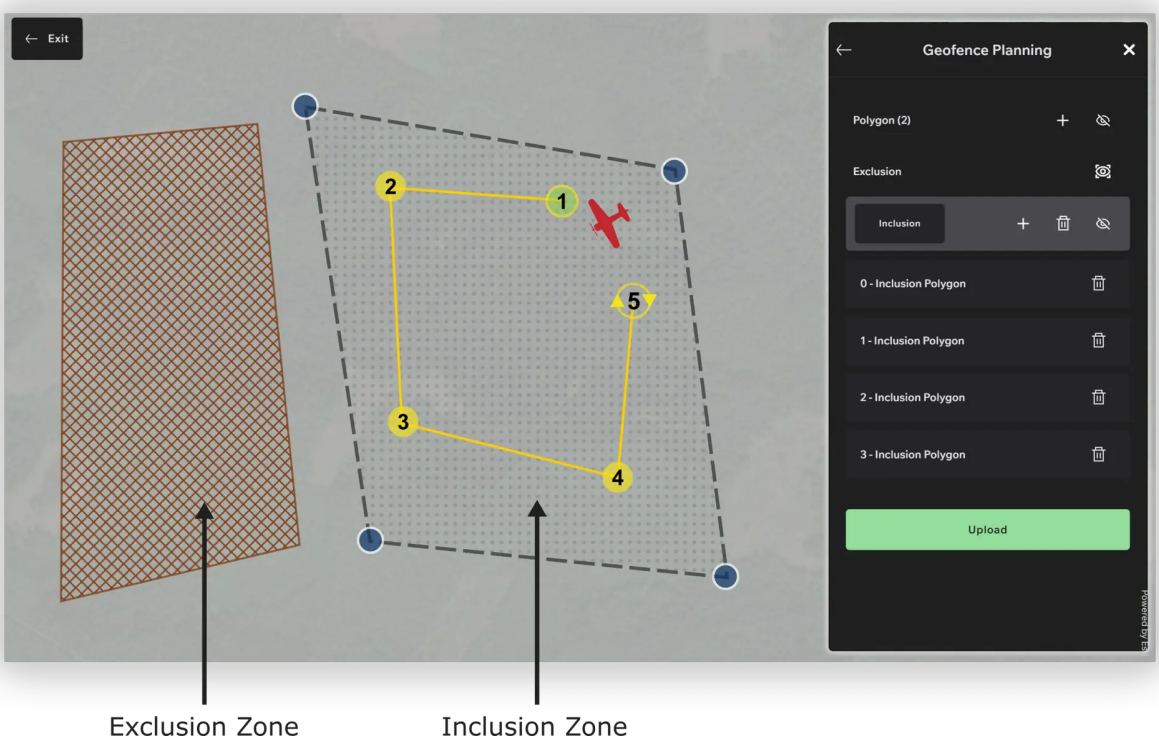


Figure 6.9: Geofence inclusion and exclusion zones

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6.7.1 Polygon Geofences

Polygon geofences are organized into groups. Each group is drawn on the map and listed in the planner panel.

To create a new polygon group:

1. Open the **Polygon** section.
2. Press the **+** button for a new group.
3. Click on the map where the new geofence should be created.

When a new polygon group is created, the planner generates an initial four-point polygon around the selected map position.

To edit a polygon group:

1. Expand the required group in the list.
2. Use the group type control to toggle between **Exclusion** and **Inclusion** zones.
    1. **Exclusion** zones define areas the UAV must not enter. Entering an exclusion zone is treated as a geofence breach.
    2. **Inclusion** zones define the permitted operating area. Leaving an inclusion zone is treated as a geofence breach.
3. Press the **+** button inside the active group to add another point, then click on the map to place it.
4. Drag the polygon handles on the map to reposition existing points.
5. Delete individual points or remove the entire group if required.

Different fill patterns are used on the map to distinguish inclusion and exclusion zones.

6.7.2 Uploading Geofences

After editing is complete:

1. Press **Upload**.
2. Use the confirmation slider to send the geofence data to the active UAV.

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7 Operation

This chapter describes the standard workflow from pre-flight inspection through arming and launch. Use the in-application **Pre-flight Checklist** as the main confirmation tool to successfully arm and launch the UAV.

7.1 Connecting to the Ground Station

Once the ground control station is prepared and catapult is assembled, connect the system as follows:

1. Power and position the UAV in the launch area.
2. Launch **SwarmC2** on the GCS tablet.
3. Wait for the software to detect the UAV, establish telemetry.
4. Confirm that the correct UAV is selected and that telemetry values are updating in the main window.

NOTE: This process can take up to 2 minutes. If connection takes longer, verify the UAV power state and radio link.

NOTE: For best radio performance, keep the UAV at least 10 meters away from the radio module while on the ground.

7.2 Pre-Flight Checklist

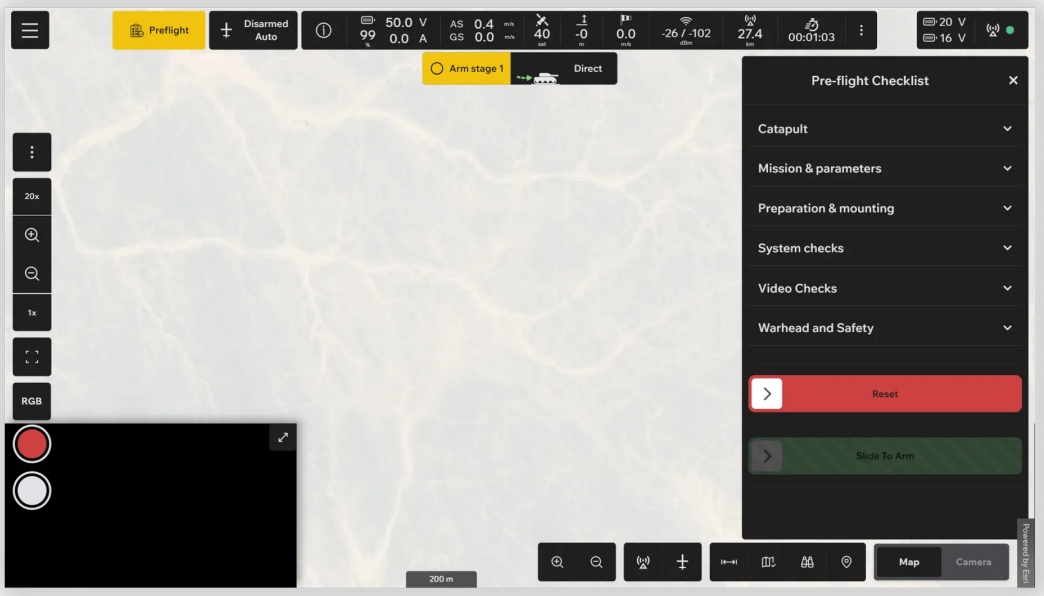


Figure 7.1: Pre-flight Checklist overview

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Before arming, complete the **Pre-flight Checklist**. The checklist combines automatically evaluated status checks with manually confirmed operator steps.

NOTE: The **Slide To Arm** control becomes available only after all required checklist items are complete.

7.2.1 Opening Pre-Flight Checklist

To open pre-flight checklist, click **Preflight** in top actions dialog

7.2.2 Checklist Behaviour

The **Pre-flight Checklist** behaves as follows:

1. The checklist opens in a pop-up window and requests the current checklist state for the connected UAV. Verification changes, overrides, flags, and resets are synchronized for the active session.
2. When the checklist is incomplete, the first incomplete group opens automatically. Only one group is expanded at a time. When the currently open group becomes complete, it collapses automatically and the next incomplete group opens.
3. Group headers indicate their current state. A group can show incomplete, warning, error, or complete status depending on the result of its checklist items.
4. Automatic checklist items are re-evaluated continuously while the UAV is connected, not armed, and not flying. These items show the automatic-check indicator and can change state without operator input.
5. Manual checklist items use checkboxes.
6. If any Pre-Arm checks fail, the Arming indicator will show **Not Ready,** when pressing the **Arm** button, will open a log which shows which systems are failing the Pre-Arm check.

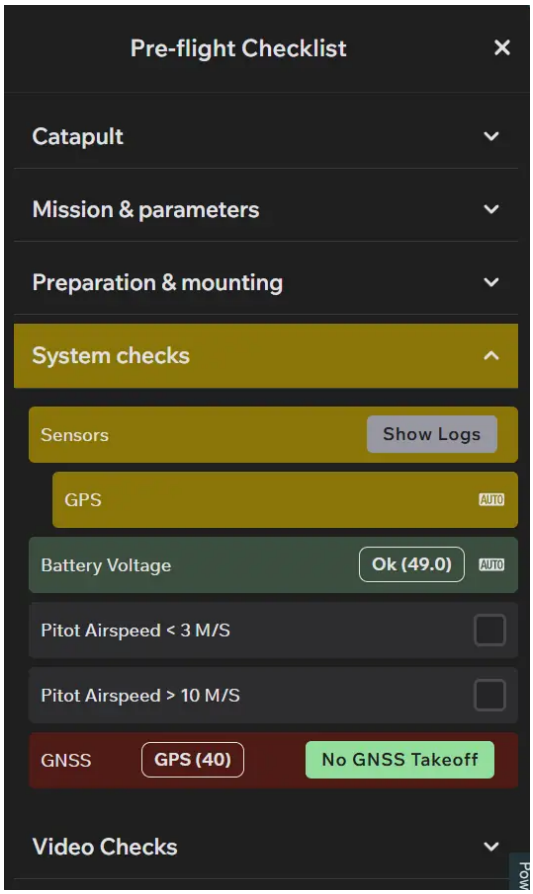


Figure 7.2: Pre flight warnings and errors

CAUTION: The pre-flight checklist may become incomplete or display an error even after successful completion if any items begin to fail afterward. If any of the items become invalid even after the checklist is complete, the program will disable **Slide to Arm** button.

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7.2.3 Checklist Structure

The current Avenger 5 pre-flight checklist is divided into six main groups.

Quick navigation:

1. *Catapult*
2. *Mission & parameters*
3. *Preparation & mounting*
4. *System checks*
5. *Video Checks*
6. *Warhead and Safety*

Catapult

This group covers the physical preparation of the catapult launch system.

1. **Assemble catapult rail** makes sure that the catapult rail is assembled and secured to the ground properly.
2. **Carriage Safety pin set** confirms that the carriage is in its initial position, secured with a **safety lock pin**.
3. **Attach pulling rope** makes sure that the pulling rope is attached to the carriage and tensioned properly.
4. **Compressor valve closed** confirms that the main reservoir outlet valve is in the **closed** position.
5. **Pressure hose connected** confirms that the pressure hose connecting main valve outlet and the launch inlet is secured, clamps are locked and safety pins installed properly.
6. **Compressor ON** confirms that the compressor battery has been inserted and the pressurization process has begun.
7. **Pressure at 13-14bar** confirms that the determined pressure for safely launching the UAV has been reached.
8. **Plane secured on catapult** confirms that the UAV has been positioned into the launch carriage.

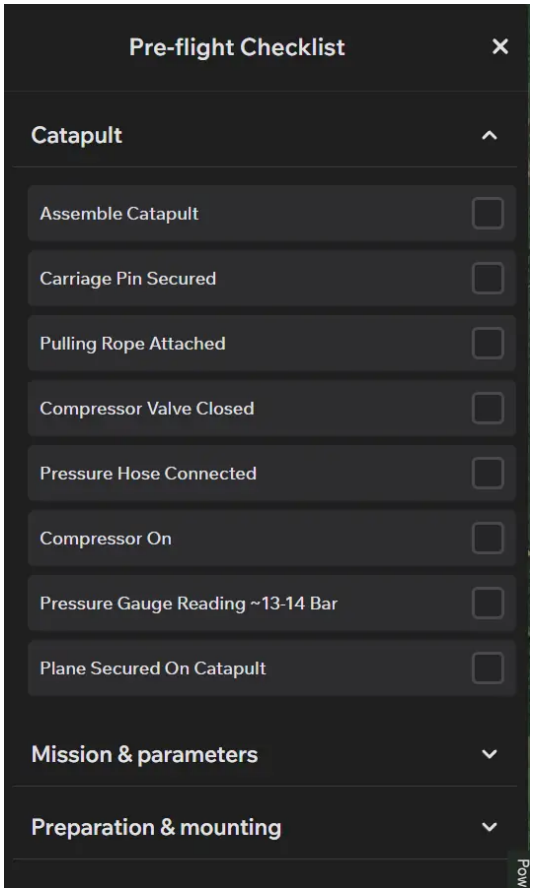


Figure 7.3: Catapult

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Mission & Parameters

This group confirms that the mission and ground-station-related flight settings are ready.

1. **Mission plan** confirms that a valid mission plan is available for the UAV. Helper button: **Mission** opens mission planning. If a mission is already loaded but the UAV is not in auto mission mode, the helper button changes to **Set mode auto**.
2. **RTL configured** confirms that the Return-to-Launch behavior has been configured.
3. **Rotator position & direction set** confirms that the antenna rotator position and pointing direction have been set and calibrated. Helper button: **Set** opens the ground-station control screen.
4. **Rotator in manual mode** confirms that the rotator is in manual mode. Helper button: **Set Manual** is shown when the rotator is in automatic tracking mode. The button sets the rotator mode to **Manual**.
5. **Antenna omni mode** confirms that the antenna is set to omnidirectional mode. Helper button: **Switch** is shown when the antenna is not currently in omni mode and changes it to **omnidirectional**.
6. **Radio power set** confirms whether Ground station radio is turned on and power is set to **2W**

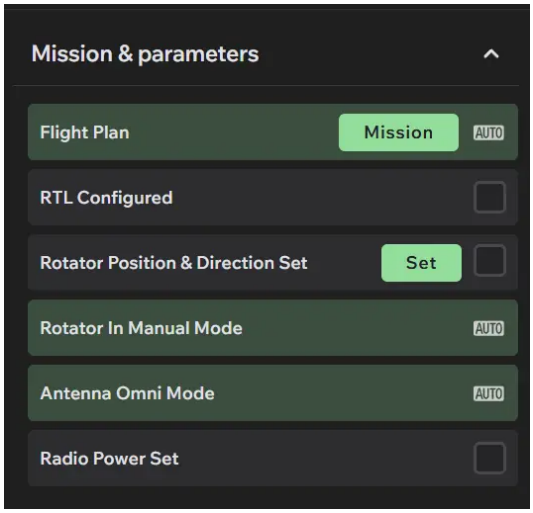


Figure 7.4: Mission & parameters

Preparation & Mounting

This group covers the physical preparation of the aircraft before launch.

1. **Stabilizers attached** confirms that the stabilizers are installed.
2. **Wings unfolded** confirms that the wings have been unfolded.
3. **Wing and stabilizer bolts tightened** confirms that the wing and stabilizer bolts are tightened.
4. **Check Propeller** inspect the propeller for any looseness or play, and tighten the screws if necessary.
5. **Servo maximum deflection checked** confirms that servo movement has been checked through full travel. Helper button: **Test** runs the servo wiggle action. The operator should visually confirm that all servos are moving correctly after pressing the helper button.
6. **Check auto stabilization** the operator must visually inspect whether all servos traveled full cycle.

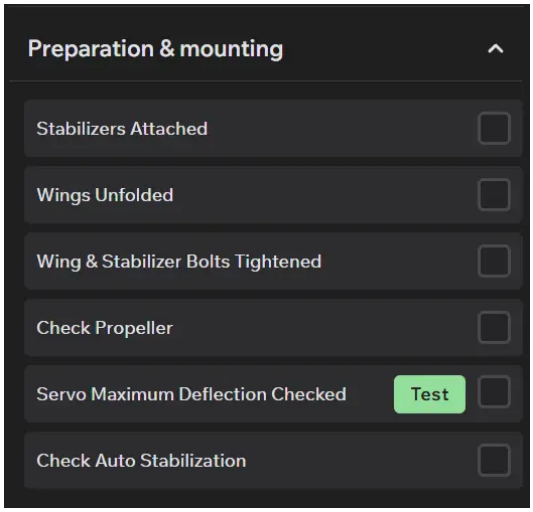


Figure 7.5: Preparation & mounting

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System Checks

This group checks aircraft health, sensor readiness, battery state, and navigation readiness.

1. **Sensors** is the parent item for the automatic sensor-health checks. Helper buttons: **Show Logs** or **Hide Logs** open or hide the sensor-error view, and **Override** appears when blocking sensor faults must be manually accepted.
2. **Battery voltage** checks that battery voltage is within the allowed pre-flight range. A fully charged battery should read approximately **49 V to 50 V**.
3. **Pitot** includes two checks:
   - **Pitot airspeed below 3 m/s** — when the aircraft is stationary and the air around the pitot tube is calm, the indicated airspeed should remain below **3 m/s**.
   - **Pitot airspeed above 10 m/s** — the operator should blow gently and indirectly into the pitot tube. During this check, the indicated airspeed should rise above **10 m/s**.
4. **GNSS** is the parent item for GNSS-related pre-flight checks. Helper button: **No GNSS Takeoff** switches the workflow to a no-GNSS launch procedure. When that mode is active, the button changes to **GNSS Takeoff**. Child checks:
   - **Disable GNSS** confirms that GNSS is disabled. Helper button: **Disable**; that action disables GPS. The item is automatically checked.
   - **Plane position set** confirms that the aircraft position has been set by the operator.

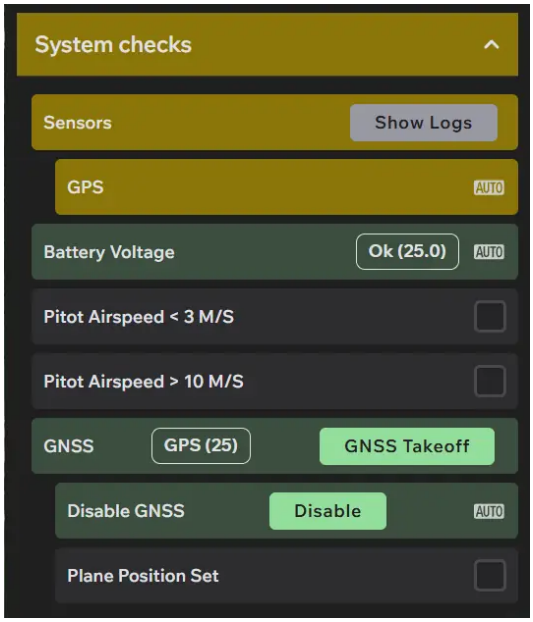


Figure 7.6: System checks

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Video Checks

This group verifies that the payload and camera system are ready for operation.

1. **Video Mode STOW** confirms that the video system can be set to STOW mode. Helper button: **Set STOW**.
2. **Video Track works** confirms that video tracking functions correctly. Clicking the video camera tracks the selected point.
3. **Video Mode GRR** confirms that the video system can be set to GRR mode. Helper button: **Set GRR**.
4. **RGB / IR Video Works** confirms that switching between RGB and IR video works. Helper button: **Toggle Camera**.
5. **IR Camera NUC works** confirms that an IR camera NUC can be performed. Helper button: **Perform NUC**.

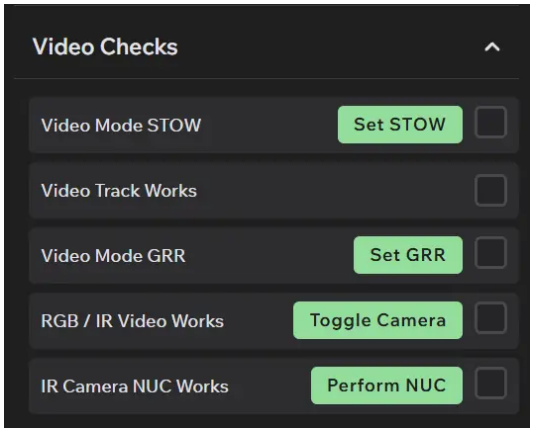


Figure 7.7: Video checks

Warhead and Safety

This group verifies that the warhead and other safety checks are ready for operation.

1. **ESAD Powered** confirms that ESAD has been powered on and is ready.
2. **Removed ESAD Safety Pin** checks whether ESAD has started and is operational.
3. **Compressor Valve Open** manual check to ensure compressor valve has been opened.
4. **Remove Carriage Safety Pin** manual check to ensure catapult safety pin has been removed.

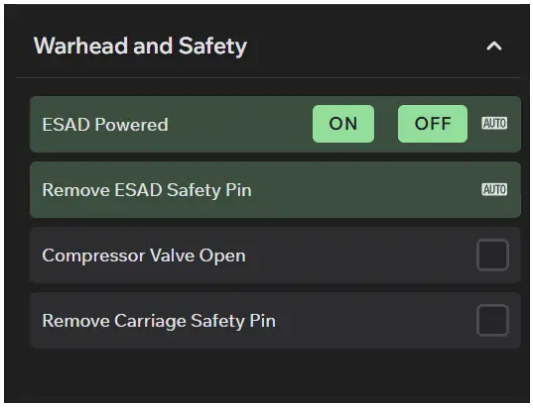


Figure 7.8: Warhead and Safety

7.3 Arming and Launching

When the **Pre-flight Checklist** is complete and no critical faults are found, the UAV is ready to be **Armed**.

To arm the UAV:

1. Open the checklist from the arm control in the flight toolbar if it is not already open.
2. Slide the **Slide To Arm** button within the pre-flight checklist panel.
3. Confirm that the UAV status in the UAV actions toolbar changes to **Armed** before proceeding with launch.

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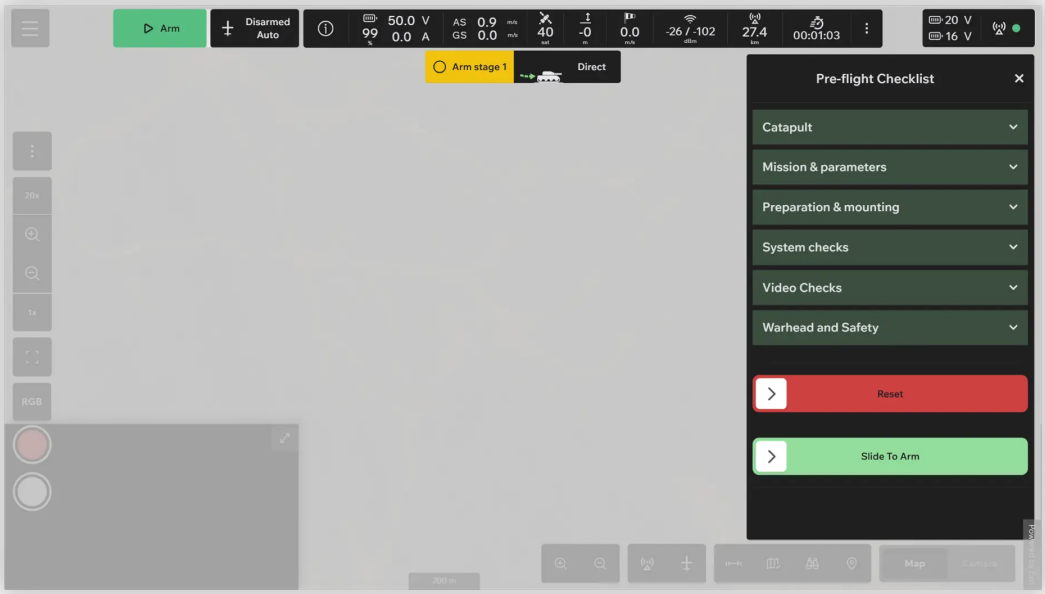


Figure 7.9: Pre-flight complete and ready to arm.

NOTE: If arming is rejected, review the **Pre-Arm Errors Found** dialog or the flight logs, correct the reported issue, and repeat the arming procedure.

7.3.1 Launch procedure

Steps for launching the UAV:

1. Move the reservoir outlet valve to the **Open** position.
2. Remove the **safety lock pin** from the carriage.
3. Clear the catapult and launch area from any personnel.
4. Press and hold the catapult trigger button untill the UAV has left the catapult.

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8 Controlling the UAV

SwarmC2 software offers various functions, which are essential for safe and effective flight using Unmanned Defense Systems UAV's. These functions include different flight modes, which are based on the currently selected platform type, controlling flight patterns directly from the main map view, as well as additional functions for direct speed and altitude adjustments.

8.1 Flight Modes

The **Avenger 5** platform has 5 directly selectable flight modes from the dropdown menu:

1. **AUTO** – UAV flies the uploaded automatic mission.
2. **CRUISE** – Vehicle maintains current heading, altitude and cruise airspeed.
3. **LOITER** – Aircraft circles around the current position that the vehicle was when entering this mode at predefined radius.
4. **RTL** – UAV returns to the **RETURN** point.
5. **MANUAL** – UAV control surfaces follow direct RC input without autopilot stabilization.

NOTE: **MANUAL** mode is only available when **Advanced** is enabled in the **Display** menu. See *Display Settings*.

To enter the desired flight mode:

1. Press flight mode button within the flight panel.
2. Select desired flight mode from the dropdown.
3. Slide to confirm.

NOTE: **GUIDED** mode is not directly selectable from the dropdown menu. This mode is entered when the UAV is controlled directly from map view using standard flight patterns.

8.1.1 Auto Mode

In **AUTO** mode, UAV follows the uploaded mission plan waypoints or mission commands sequentially, based on the mission item number.

When flying towards the next mission item, vehicle will automatically control its direction and altitude, while maintaining **Cruise airspeed** unless specified otherwise by the operator. Vehicle will always choose the most direct path towards the next mission item and will set its desired climb or descent rate so that the target altitude is reached at the target location. Waypoints can be completed regardless of current altitude, even if the target altitude has not yet been reached.

NOTE: If UAV is flying towards **Loiter to Altitude** mission item, vehicle will only complete this item when both target **position** and **altitude** is reached.

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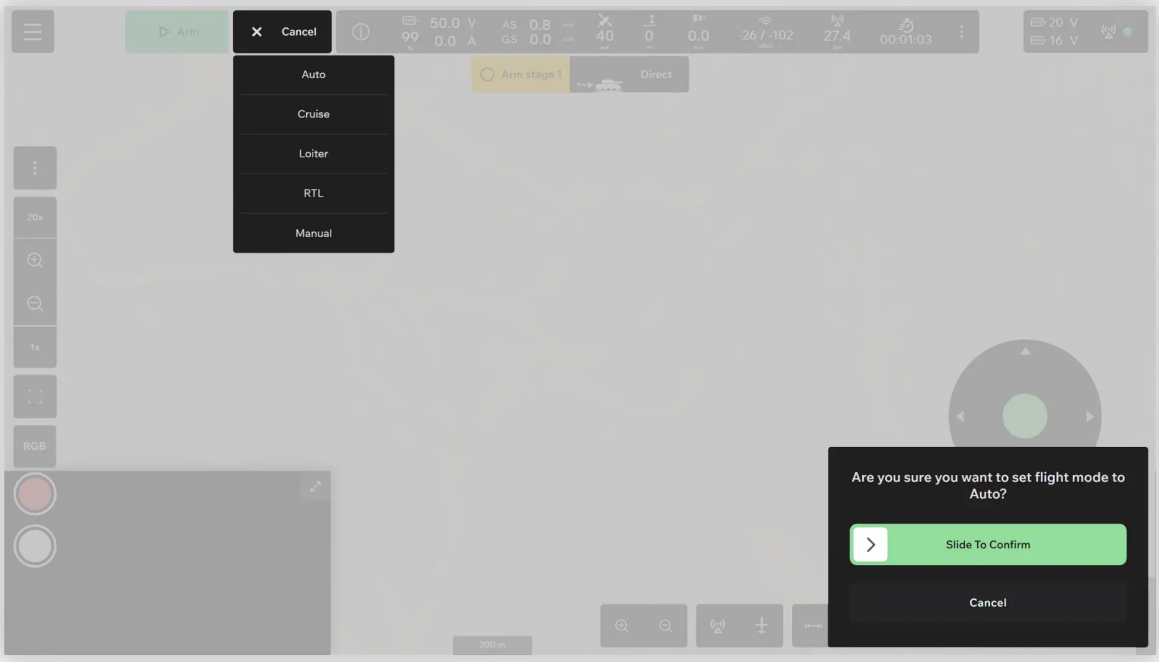


Figure 8.1: Changing flight mode.

The mission can be paused at any point by simply changing the flight mode, setting a new flight pattern from the map function menu, or dragging an existing flight pattern already present on the map. To resume the mission from the last waypoint, change the mode back into **AUTO**.

NOTE: If there is no mission plan present, vehicle will not allow to enter **AUTO** mode, and will default to **RTL**.

The mission plan is always visualized on the map for the currently selected aircraft, and can have different statuses depending on the progress of the mission:

1. **Takeoff / Home point**
   It's always marked as the first point in the mission and is automatically updated to the latest GNSS position before **Arming** the vehicle. This point is the default return point for **RTL**

2. **Completed mission item**
   Marks mission items that have already been completed or skipped by the operator.

3. **Target mission item**
   Marks the mission item that is currently being executed. When re-entering **AUTO** mode, vehicle will continue executing the mission from this point.

4. **Incomplete mission item**
   These items have not been reached or executed, and remain pending in the mission sequence.

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8.1.2 Cruise Mode

In **CRUISE** mode, UAV is maneuvered with the use of a joystick, and when no joystick input is present, maintains its current heading, altitude and **Cruise airspeed**. The joystick controls are not direct control surface inputs, but demands for climb rate and turn rate, so there is no risk of running the vehicle out of control due to sharp maneuvers.

NOTE: The virtual joystick is only visible in the **Pilot** layout, and when UAV is in **CRUISE** mode, at the bottom right of the main map view.

The joystick has standard axis for plane control:

- **UP** – Decrease altitude.
- **DOWN** – Increase altitude.
- **LEFT** – Turn left.
- **RIGHT** – Turn right.

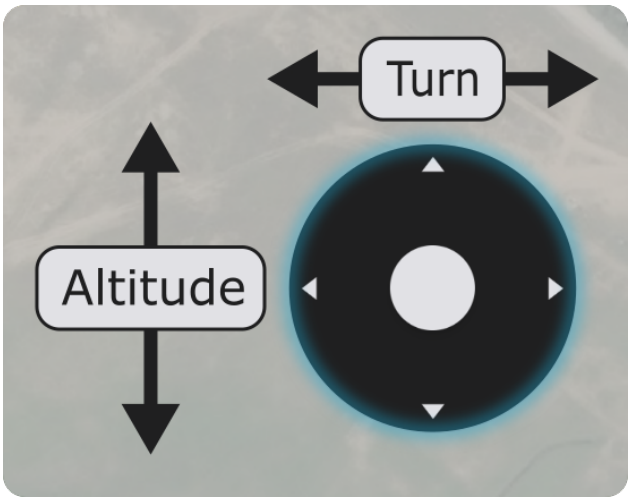


Figure 8.2: Pilot joystick control axis

CAUTION: Use caution when flying low to the ground in **CRUISE** mode, as vehicle controls altitude like in other modes, and takes some time to respond to joystick altitude inputs.

8.1.3 Loiter Mode

In **LOITER** mode, the UAV circles around the position where it was when the mode was activated, while maintaining its current altitude and **Cruise airspeed**.

The loiter direction is always set **clockwise**, and the radius of the circle is based on the platform that is currently being used:

- Avenger 5 - 300m

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8.1.4 RTL Mode

In **RTL (Return-to-Launch)** mode, the UAV will return to its home location (usually set as the location the UAV was **Armed**) and loiter there indefinitely, or execute alternate instructions as configured inside the vehicle's **Safety** menu. If rally points are defined in the uploaded mission plan, the autopilot may divert to the nearest rally point instead when it is closer and within range. See *Rally Points*.

For **Avenger 5** loitering munition platform, home location is set as the **Return Point**.

Depending on the **operational environment**, different RTL behavior for altitude control or actions may be required for safe UAV operation where loss of radio link is expected:

1. Altitude control:
    - **Return at Current Altitude** - The UAV flies back to its home location at its current altitude when entering **RTL** mode.
    - **Return at Specified Altitude** - The UAV flies back to its home location, while also climbing/descending to the configured **RTL Altitude**.
2. RTL Action:
    - **Disable** - RTL functionality will not be triggered after communication is lost. Vehicle will proceed with the auto mission or continue executing the last command.
    - **Fly HOME then Land** - Vehicle will return to **HOME** point first, then proceed with the automatic landing sequence. While flying towards **HOME** point, vehicle will be in **RTL** mode and maintain configured altitude control behavior.
    - **Go Directly to landing sequence** - Vehicle will fly straight towards the automatic landing sequence in **AUTO** mode, gradually descending towards landing altitude.
    - **Only For Go Around** - Vehicle will return to **HOME** point and then enter a standard loiter pattern at configured **RTL** altitude.

CAUTION: Disabling **RTL** functionality significantly increases the risk of losing and damaging the UAV, especially when flying in **CRUISE** or **GUIDED** modes.

8.1.5 Manual Mode

In **MANUAL** mode, the autopilot passes pilot stick inputs from the RC link directly to the control surfaces. The vehicle does not stabilize heading, altitude, or airspeed.

Unlike **CRUISE** mode, SwarmC2 does not provide on-screen joystick or gamepad flight control in **MANUAL** mode. Use the aircraft RC transmitter for direct control.

CAUTION: **MANUAL** mode is intended for experienced operators only. Loss of RC link in **MANUAL** mode triggers standard RC failsafe behavior configured in the vehicle **Safety** menu.

8.1.6 Guided Mode

When selecting **Loiter Here** map functionality, a pattern will appear on the map, which allows for quick and easy navigation towards a desired location, without the use of a mission plan.

To set a new loiter pattern on the map, or change the existing pattern's flight altitude:

1. Tap on the map, at the desired pattern location.
2. Select **Loiter Here** from the context menu.

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3. Adjust location and dimensions by dragging the appropriate icons on the displayed flight pattern on the map.
4. Adjust pattern flight altitude using the altitude slider or input buttons.
5. Slide to confirm.



Figure 8.3: Set new loiter pattern

NOTE: After a flight pattern has been set, its position and flight radius can be changed by dragging the icons found on the displayed flight pattern on the map. When the loiter pattern is updated by dragging one of the pattern icons on the map, the UAV will immediately return to **GUIDED** mode if currently flying in other modes, and start navigating towards the new pattern.

8.1.7 Attack Mode

This mode is entered when operator decides to engage a tracked target through the video feed. When UAV is in **ATTACK** mode, it will guide itself automatically towards the tracked target, while removing any control constraints for attitude or airspeed present in other modes.

To enter **ATTACK** mode:

1. Press **ENGAGE** button within the flight panel.
2. Slide to confirm.

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NOTE: To enter **ATTACK** mode, a valid track needs to be set and the warhead must be in the **ARMED** state.

8.2 Additional Vehicle Commands

In addition to the main UAV controls, additional commands for controlling flight altitude and airspeed are available in **AUTO**, **GUIDED** and **LOITER** modes. These commands temporarily override the current target airspeed and/or altitude, and will return back to the previous value after:

- Loiter pattern is updated on the map.
- UAV reaches a waypoint while executing an automatic mission plan.
- Vehicle enters an automatic landing sequence.
- Vehicle changes modes.

CAUTION: Make sure to plan for altitude or airspeed changes in advance, as changing altitude frequently or flying at high airspeeds severely increases power consumption and will reduce maximum flight time.

8.2.1 Change Altitude

To change current UAV target altitude:

1. Tap the altitude telemetry box on the instrument panel.
2. Use the dropdown panel to select target altitude.
3. Press confirm.

NOTE: When the target altitude is set through this panel, UAV will try to reach it as fast as possible, at the maximum available ascent/descent rate.

8.2.2 Change Airspeed

To change current UAV target airspeed:

1. Tap the aircraft speed telemetry box on the instrument panel.
2. Use the dropdown panel to select target airspeed.
3. Press confirm.

NOTE: When UAV enters **RTL** mode, if the current target airspeed is set above cruise speed, UAV will maintain that speed until the vehicle leaves **RTL** mode.

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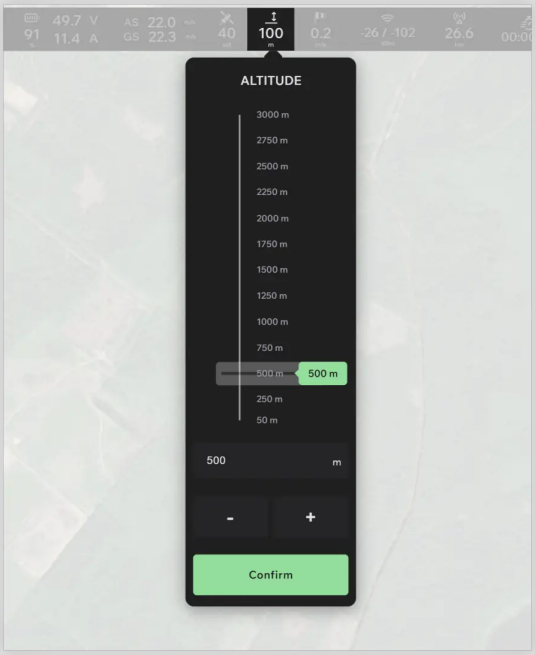


Figure 8.4: Change altitude panel

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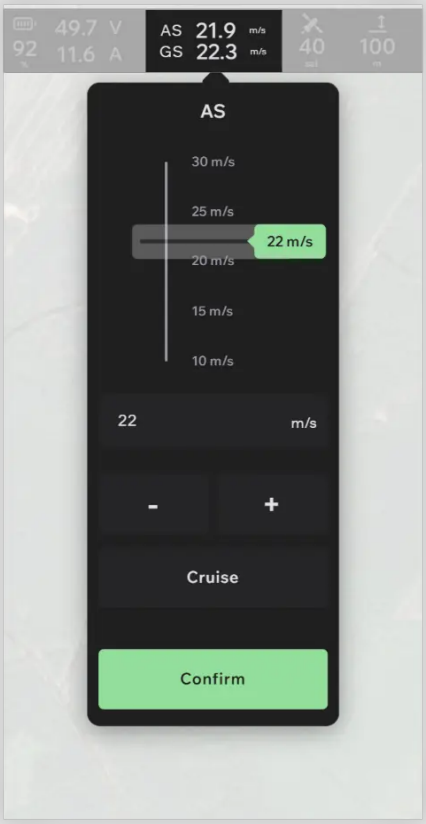


Figure 8.5: Change airspeed panel

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9 Map Operations

9.1 Five Whiskey Report

SwarmC2 software allows the user to add, edit or remove 5W reports from the map, which can greatly increase situational awareness when utilizing external tactical awareness software like **SitaWare** or **ATAK**.

NOTE: All reports from **SwarmC2** software are automatically synced, if connected to external tactical awareness software.

To add a report to the map:

1. Press on the map.
2. Select **Five W**.
3. Complete the fields so the report carries **5W** (who, what, when, where, why) information for the map and any linked tactical clients:

   Unit – Unit function or type (for example infantry or reconnaissance). This sets the symbol on the map.

   Standard Identity – Whether the contact is friendly, hostile, neutral, or unknown.

   Dimension – Operating environment (for example ground, air, or sea).

   Status – Whether the unit is **present** at the location or **anticipated**.

   Team Size – Echelon (for example squad, platoon, or company), or unknown.

   Movement – Stationary or direction of movement.

   Identified Unit – Name, call sign, or designation of the specific force (required).

   Comment – Optional remarks or context.

   Where are the coordinates of your map press; **When** is applied automatically when you submit.
4. Press **Mark unit.**

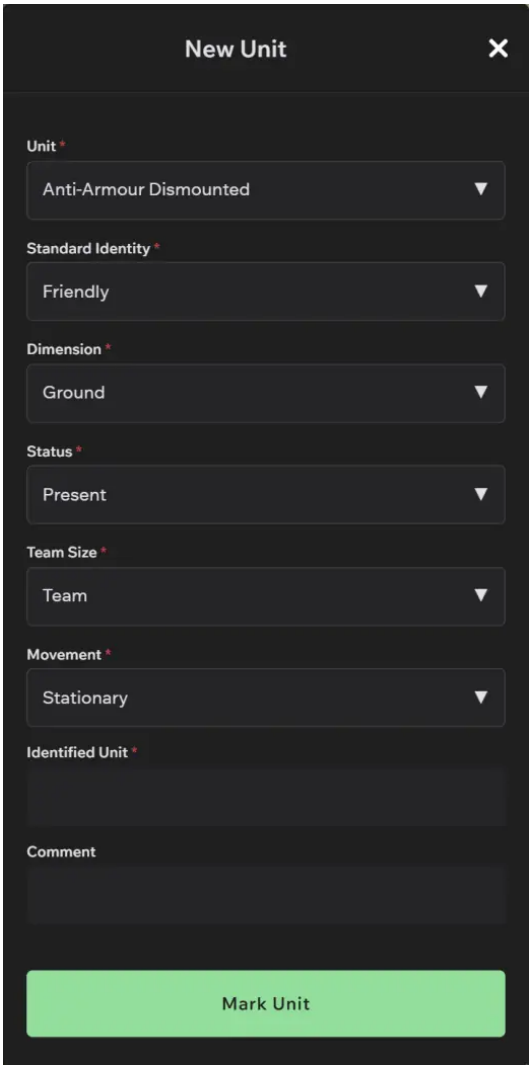


Figure 9.1: Five Whiskey report

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9.1.1 Edit Report

To move an existing 5W report:

1. Press and hold on the 5W icon on the map.
2. Drag towards and release over a desired location.
3. Slide to confirm.

To delete or perform a BDA on an existing 5W report:

1. Press on the 5W icon you want to edit or perform BDA.
2. Select **Edit Five W / BDA** from the function menu.
   a. Press **delete** to remove 5W report from the map.
   b. Add appropriate information and press **Submit BDA**.

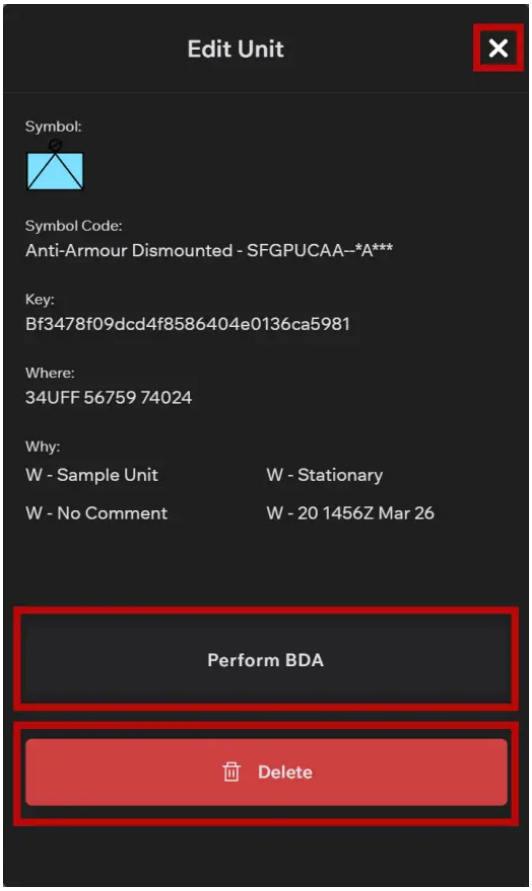


Figure 9.2: Edit Five W / BDA

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9.2 Call For Fire

This feature allows the user to send a CFF request message to external tactical awareness software, like **SitaWare** or **ATAK**. To send a CFF request:

1. Press on desired coordinates on the map.
2. Select **Call For Fire Request** from the function menu.
3. Fill out the required information.
4. Slide to confirm.

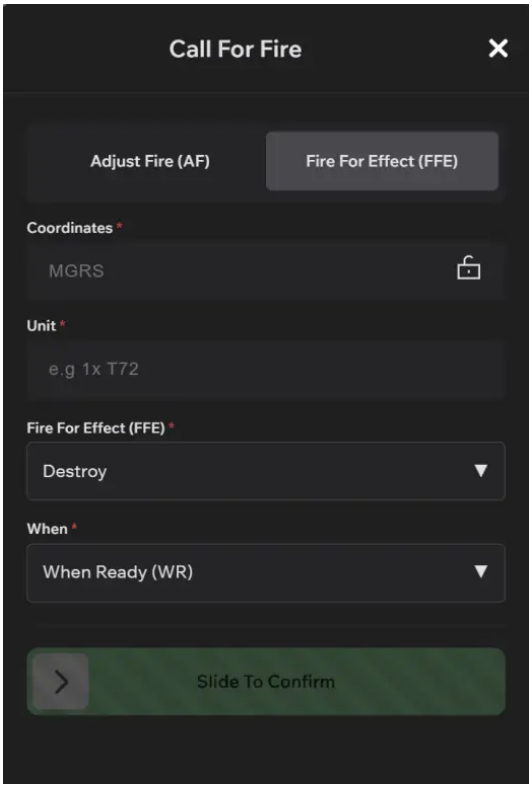


Figure 9.3: Call For Fire Request

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9.3 Calculate Distance between 2 Points

To calculate linear distance between two points:

1. Press on *Ruler* icon
2. Ruler mode will be enabled
3. Press on the map where you want to calculate distance from
4. Press on the map where you want to calculate distance to
5. You can press and drag any point to move it to the desired location
6. Total distance is shown in top center



Figure 9.4: Total distance

9.4 Points of Interest

Points of interest are a separate layer of markers which help operators to use supported UAVs efficiently. These markers are stored locally and are separate from 5W reports and mission plans.

To add a new POI marker:

1. Press on desired coordinates on the map.
2. Select **Add Point of Interest** from the function menu.

To edit POI marker:

1. Press on a POI you want to edit.
2. Select **Edit Point of Interest** from the function menu.
3. Select POI type and desired options.
4. Press confirm.

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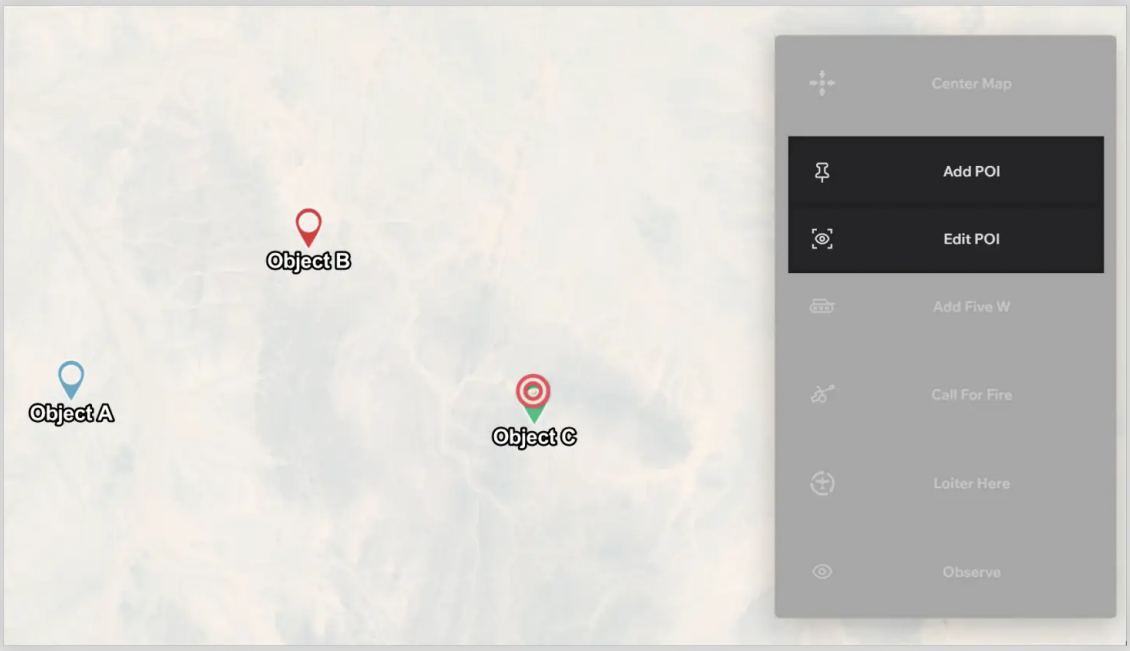


Figure 9.5: POI markers

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10 Radio Controls

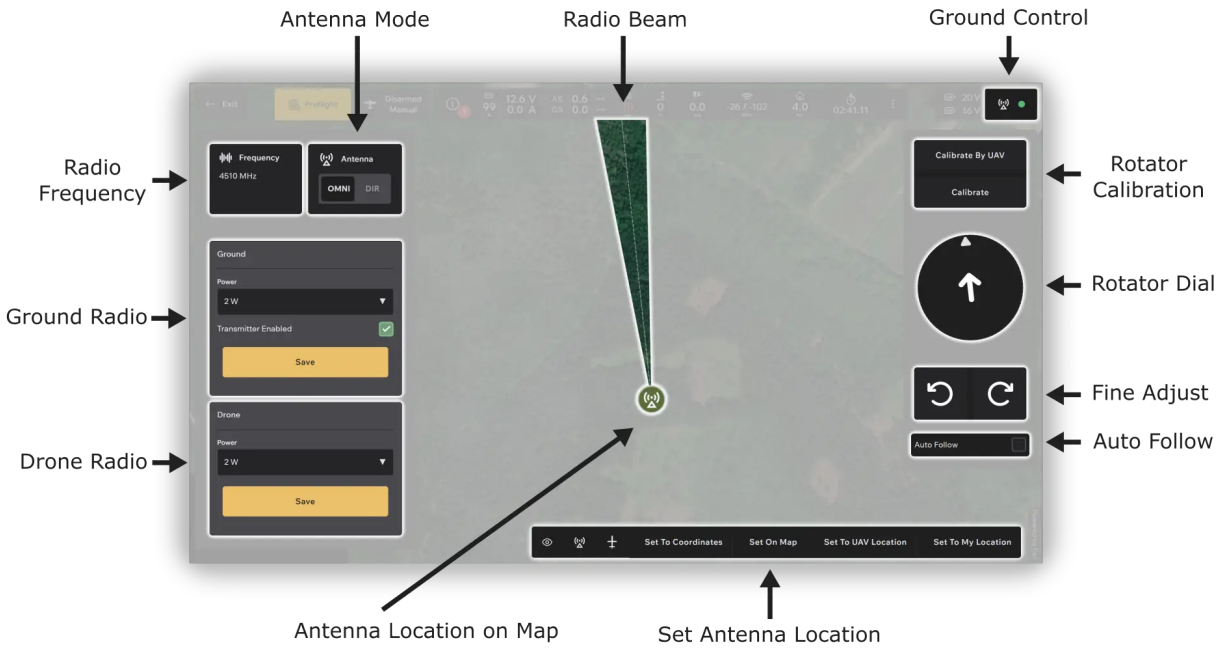


Figure 10.1: Rotator Control Interface

GCS control panel features a **Rotator Control Interface,** which is opened by pressing the **GCS** button, and features both radio configuration, and rotator controls:

Rotator control interface button
Opens the Ground Control view

Current radio frequency
Shows currently selected UAV's Uplink/Downlink frequencies. If no vehicle is connected, then it shows only GCS frequency.

Antenna toggle
Switch between omni-directional and directional antennas, that GCS will be using for communication.

Ground Radio power control
Set currently selected GCS power output. With Enable/Disable ground transmitter radio output.

Drone Radio power control
Set currently selected vehicle power output.

Set rotator location
Sets GCS rotator location on the map. swarmC2 allows multiple ways of setting rotator location:

NOTE: When rotator is placed on the map, it is possible to drag rotator icon on the map to fine-tune its location.

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Rotator calibration
Recalibrates rotator direction to:

Manual rotator control
Sets rotator direction to a specified heading.

Fine rotator control
moves rotator by 0.5° towards a selected direction.

Auto Follow toggle
Enables/Disables auto follow mode.

10.1 Set antenna location

Use one of the location buttons to set the antenna location used for tracking and alignment.

Set To Coordinates
sets antenna location to specified LAT/LON or MGRS coordinates

Set On Map
sets antenna location to current map view's screen center.

Set To UAV Location
sets antenna location to currently selected UAV location.

Set To My Location
sets antenna location to current computer's location, if available.

10.2 Antenna Rotator Map calibration

Aligns the antenna direction on the map only. The antenna does not rotate physically.

Calibrate By UAV
calibrates rotator position to last UAV position.

Calibrate
calibrates rotator position towards a specified map location.

Before flight, set the rotator position on the map and calibrate its heading to achieve the best radio communication performance.

10.2.1 Pre-flight Rotator map calibration procedure

1. Set the rotator position using one of the available functions to achieve the highest possible accuracy.
2. Physically set the rotator direction:
    - Toward a point of interest visible on the map.
    - Toward a specific heading.
    - Toward north.

NOTE: When the rotator location is set, the initial rotator heading is north.

3. Use **Calibrate On Map** to align the software direction with the physical rotator heading set in the previous step.

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CAUTION: When using **Auto Follow** mode, make sure the vehicle is not loitering around the rotator location. Multiple rotations increase the risk of tangling or damaging GCS cables.

CAUTION: If multiple UAVs are connected at the same time, rotator **Auto Follow** tracks the **currently selected** UAV.

10.3 Antenna In-Flight calibration

This calibration applies a physical rotator movement, so the antenna rotates to the corrected heading. Use **Fine Tune** during flight, when the UAV is airborne and **Auto Follow** is active, to refine antenna pointing.

10.3.1 In-flight fine tune calibration procedure

While **Auto Follow** mode is enabled, use **Fine Tuning** mode to adjust the rotator manually.

In **Fine Tuning** mode make small heading corrections and use the RSSI value as guidance.

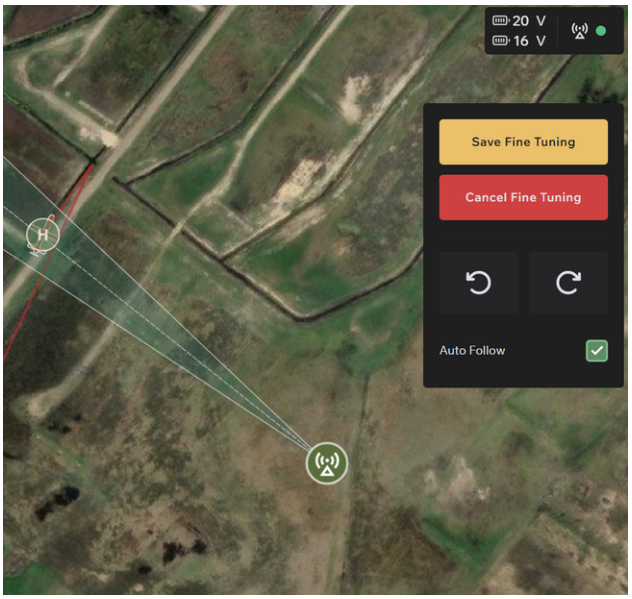


Figure 10.2: Rotator Fine Tune

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10.3.2 Rotator direction indicators

When changing rotator direction manually, or when using **Auto Follow** mode, the main map view shows:

1. **True Direction** – directly measured by the rotation mechanism inside the GCS rotator module.
2. **Desired direction** – software-defined target rotator heading.



Figure 10.3: Rotator visualization

CAUTION: When flying with **disabled GNSS** for navigation and **Auto Follow** enabled, the rotator follows the navigational position even if a valid GNSS position is present and shown on the map with a yellow aircraft icon.

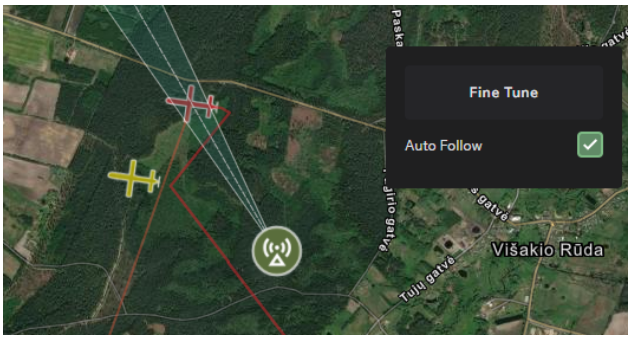


Figure 10.4: Rotator Auto Follow without GNSS

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11 Payload Controls

Camera payload is controlled directly from the main window by using the **Payload control panel,** which can be found at the bottom left edge of the screen, as well as touch functionality from the map or video feed.

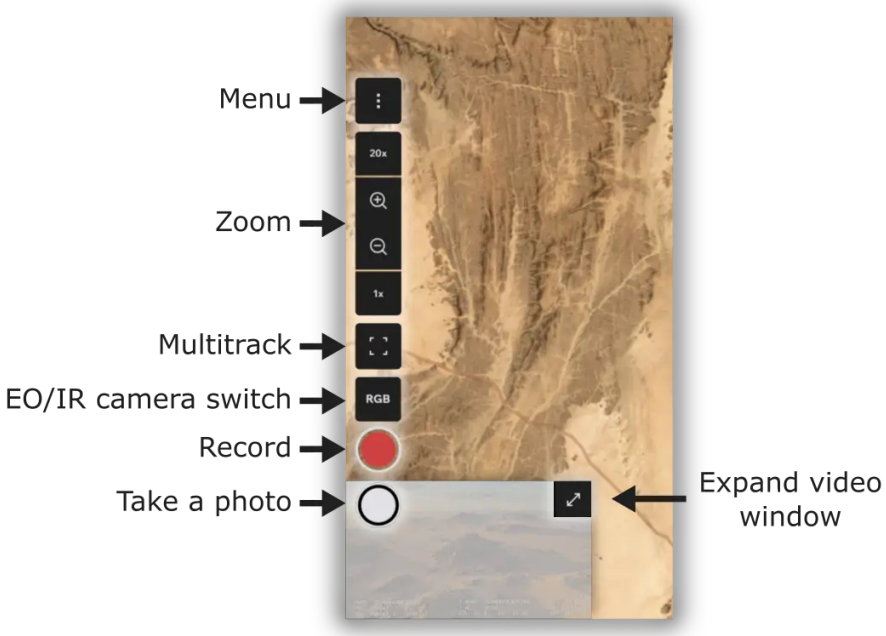


Figure 11.1: Payload control panel

Payload control panel includes the most used payload controls:

Menu
Payload configuration menu

Zoom
Controls the zoom level of the video stream. Four buttons on the payload bar:

20× and **1×** jump to high and unity zoom

+ and **−** step zoom in and out.

Multitrack
Payload capability for maintaining and managing multiple tracked objects (if available on current payload build).

EO/IR camera switch
Switches between EO (Daylight) and IR (Thermal/night) camera video stream.

Record
Starts/stops recording of the video stream. Video is saved to the payload SD card.

Take a photo
Takes a high-quality photo and saves it to the payload SD card.

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Expand video window
Drag across the screen to resize the camera view box larger or smaller.

11.1 Camera View

The main layout shows the live EO or IR video from the payload.

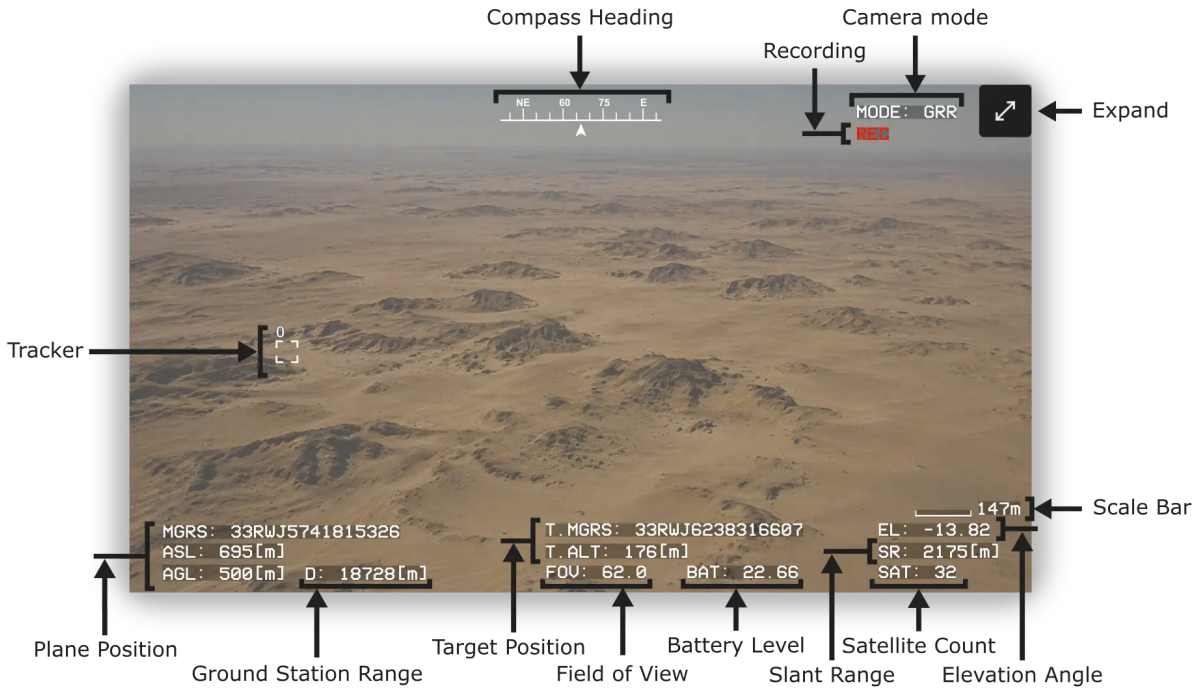


Figure 11.2: Camera

The **information layer** on top of the feed includes:

Compass Heading
Scale aligned with camera yaw so you can read direction against the horizon.

Recording
Shows when the feed is being recorded

Camera Mode
Active camera mode

Tracker Frame
The on-video box that shows what the current tracker is following.

Plane position
Coordinate readouts in MGRS, altitude above sea level (ASL), and above ground (AGL).

Target position
Target coordinate readouts in MGRS.

Ground Station Range
Distance to Ground Station

Elevation angle
How far above or below the horizon the camera is looking: the pitch of the line of sight.

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Field of View
How wide an area the camera captures

Battery level
Remaining charge from aircraft or payload telemetry.

Slant range
Straight-line distance from the aircraft to the point you are looking at.

Satellite count
How many GNSS satellites are in use for the position fix.

Scale bar
A labeled line segment whose length matches a ground distance at the current view and geometry.

Expand
Control at the top right of the video; drag it to resize the camera view

11.2 Payload configuration

Open **Menu** on the payload control panel to reach the camera configuration menu. From there you can extend or retract the gimbal, set camera orientation, choose how EO and IR are displayed, and adjust IR image options.

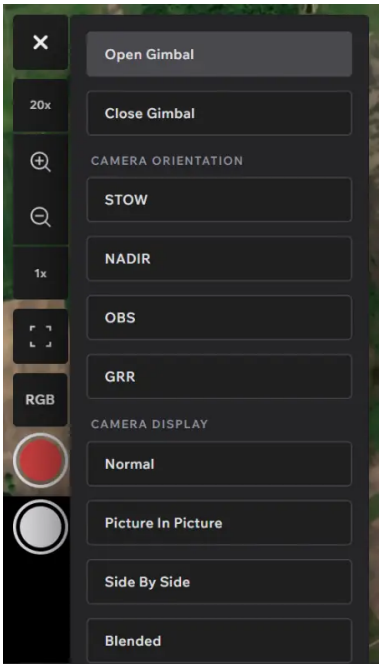


Figure 11.3: Camera configuration menu

NUC
Recalibrates IR camera to adjust for current temperature.

NOTE: NUC should be used right after takeoff or when changing altitude to adjust for temperature differences.

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11.2.1 Camera mode

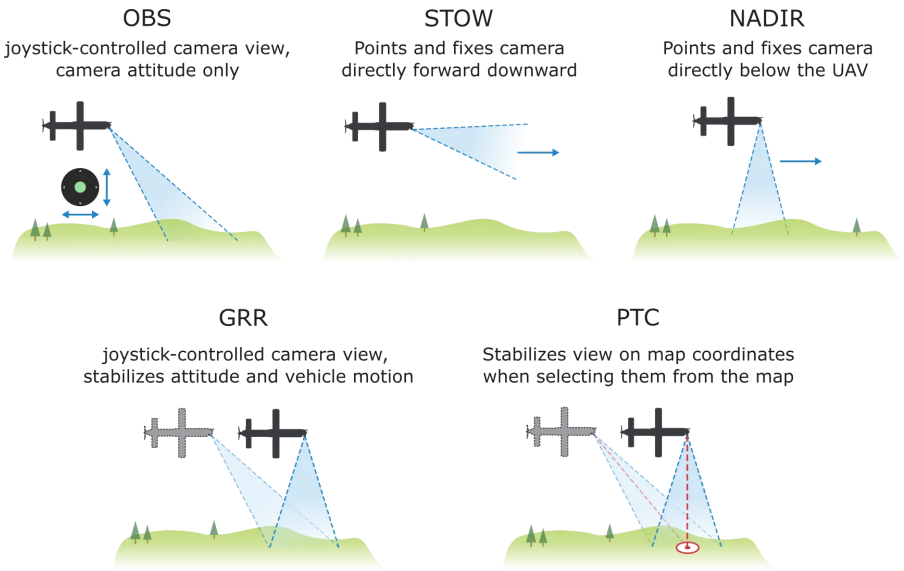


Figure 11.4: Camera Modes

STOW
Points and fixes camera directly forward downward.

NADIR
Points and fixes camera directly below the UAV.

PTC
Stabilizes view on map coordinates when selecting them from the map. To use this, tap or press the map and select **Observe** from the function menu.

OBS
Camera view follows the platform's motion.

NOTE: Joystick control is not possible when in OBS mode. Make sure to switch out of OBS mode to continue using the joystick.

GRR
Is controlled by moving the joystick and stabilizes view on map coordinates when not moving the joystick.

11.2.2 Camera Display

Normal
Shows a single full-frame video feed from the active sensor (EO or IR).

Picture in Picture
Shows the primary sensor full screen with the other sensor in a smaller inset window, so both views are visible at once.

Side by Side
Splits the video area and shows EO and IR streams next to each other at the same time.

Blended
Fuses EO and IR imagery into one combined view (payload-dependent), for a single stream that blends visible and thermal detail.

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11.2.3 IR Color Mode

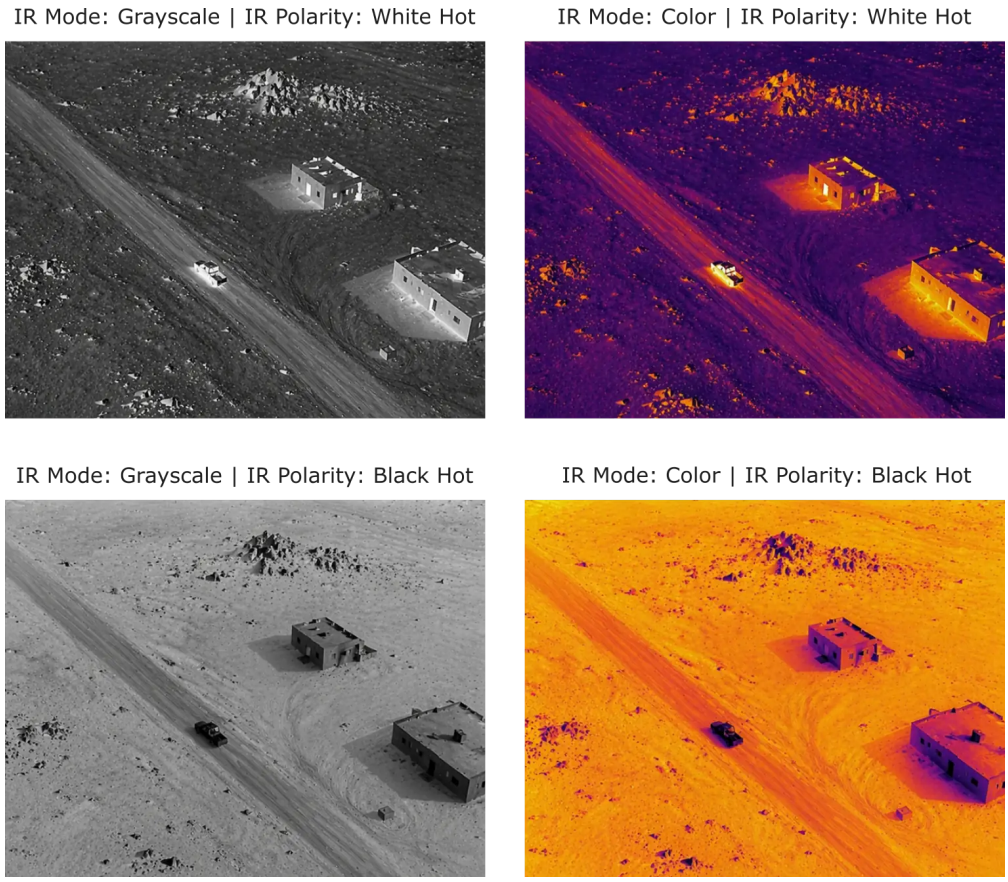


Figure 11.5: IR color modes

Grayscale
Renders the IR image in shades of gray (typical thermal view).

Color
Applies a false-color palette to the IR image to emphasize temperature differences.

11.2.4 IR Polarity

White Hot
Warmer objects appear brighter (light) against a darker background.

Black Hot
Warmer objects appear darker against a brighter background.

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11.3 Camera Control

You can control the camera in three ways: on the **live video**, on the **map**, or with a **joystick** (on-screen or USB controller). **Camera mode** defines how the payload interprets those inputs.

11.3.1 Live video

Tap or click an object in the stream to point the gimbal at it and keep it centered. When multitrack is enabled, that applies to the **active tracker** selected in the multitrack sidebar (see **Multitrack** below).

11.3.2 Map

When in OBS mode you can press on map and select **Observe** from the menu



Observe – Points the camera payload toward the selected coordinates and maintains continuous track.

11.3.3 Joystick (gimbal)

CAUTION: To use the **on-screen** joystick for gimbal control, select **Payload layout**. In **Pilot layout**, the same control is used for manual aircraft control in **Cruise** mode, not for the gimbal.

Virtual joystick

In **Payload layout**, the **virtual joystick panel** drives gimbal pan (left/right) and tilt (up/down). Touch or press on the joystick, then drag; releasing recenters the control and stops motion.

USB controller (optional)

When a compatible USB game controller is connected, **Main menu** shows **Joystick**. Open **Joystick Keys Mapping** and turn on **Enable / Disable Joystick**. You can **bind buttons**: for each listed action (camera modes, zoom, track, photo, recording, gimbal open/close, and similar), select the bind control and press the controller button you want to use; use the delete control beside that row to clear a button binding. **Axis Left** and **Axis Right** are assigned separately to **Gimbal** or **UAV Control** (defaults: left **Gimbal**, right **UAV Control**).

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11.4 Multitrack

Multitrack lets the payload maintain several simultaneous point tracks when the aircraft and payload build support it (see **Multitrack** on the payload control panel above).



Figure 11.6: Multitrack

Opening the menu
Tap the track icon on the payload control bar to open or close the multitrack sidebar.

Active tracker
Assign a tracker slot (0–4) to individual objects on screen.

Primary tracker
Set which tracker slot (0–4) the payload treats as the primary tracked object.

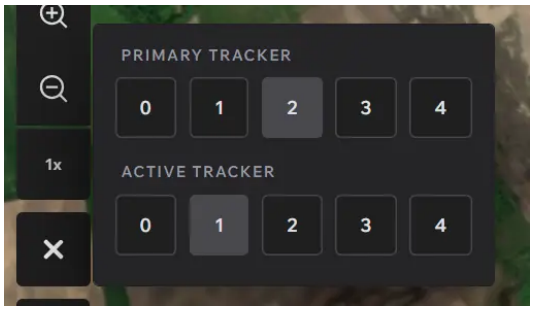


Figure 11.7: Payload Multitrack

11.4.1 Example: assigning and promoting a tracker

To assign an object to **slot 0** and set it as the primary tracker:

1. Set the active tracker to **slot 0**, then tap the desired object on the video stream. The number **0** appears over the tapped point, confirming the assignment.
2. Set the primary tracker to **slot 0**. The camera will begin following the object assigned to that slot.

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12 Target Engagement

Avenger 5 Loitering munition platform features a fully automatic guidance system capable of engaging both stationary and moving targets, which are tracked by the gimballed EO/IR camera sensor.

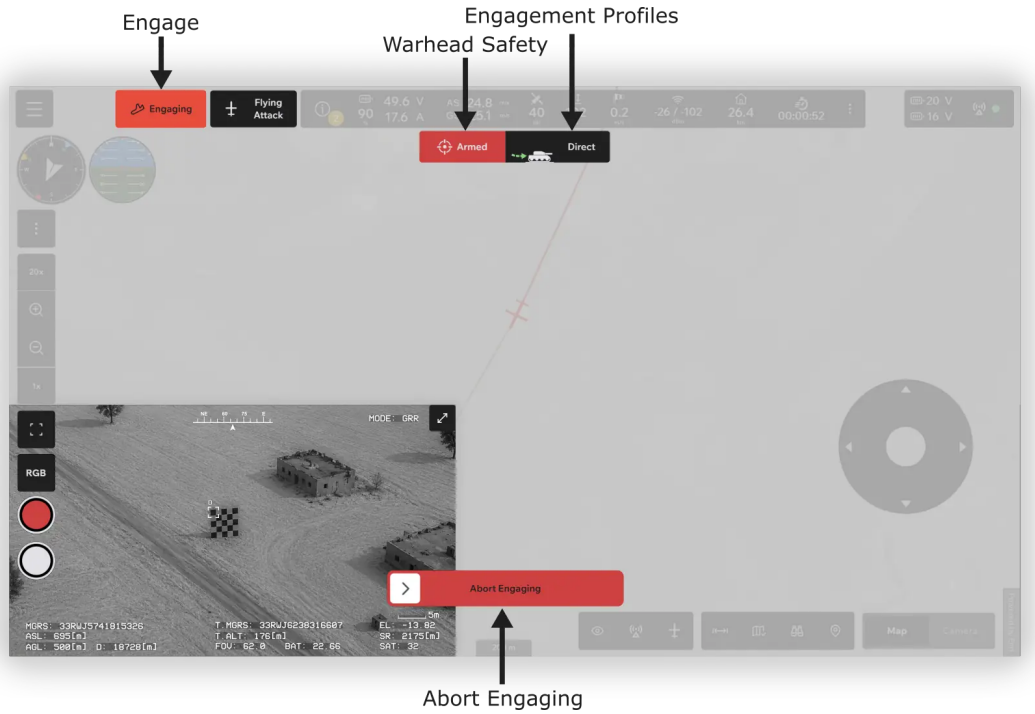


Figure 12.1: engagement and warhead controls

Interface elements for engagement and warhead control can be found in the **Main window**:

1. **Engage** – Initiates automatic guidance towards the tracked target.
2. **Warhead Safety** – Includes controls for operating the ESAD.
3. **Engagement Profiles** – Switches the engagement profile which the UAV flies during terminal guidance.
4. **Abort Engaging** – Initiates an automatic wave-off sequence away from the target.

12.1 Warhead Safety

The Electronic Safe and Arm Device (ESAD) is the principal safety device that prevents accidental warhead detonation and requires a proper sequence of events and certain conditions to be met before the warhead can be armed.

Functionally the device has 6 states, which are always visible in the **Main window**:

- **INACTIVE** - Device is powered off.
- **SAFE** - Device is powered on, but the safety pin is still in place.
- **ARM STAGE 1** - Safety pin is removed.

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- **ARM STAGE 2** - Device is airborne and within a safe distance from launch point.
- **ARMED** - Device is armed and ready. This state is initiated by pressing a button in SC2 software.
- **FIRED** - Device has been detonated by proximity sensor trigger or impact.

The operator is allowed to **ARM** the warhead when all of the described conditions are met:

1. Device is powered and no faults are detected.
2. Safety pin is removed.
3. UAV has successfully completed a takeoff.
4. UAV is currently in flight.

After all of the conditions are met, the operator can **ARM** or **DISARM** the warhead:

1. Press **Warhead Safety** status icon.
2. Press **ARM / DISARM**.
3. Slide to confirm.

The user is also allowed to self-destruct the device by following the steps below:

1. Press and hold **SELF DESTRUCT** for 5 seconds.
2. Slide to confirm.

The user is also allowed to Abort the mission by following the steps below:

1. Press and hold **ABORT** button for 5 seconds.
2. Slide to confirm.

NOTE: After aborting the mission, there is no way to turn the device back on.

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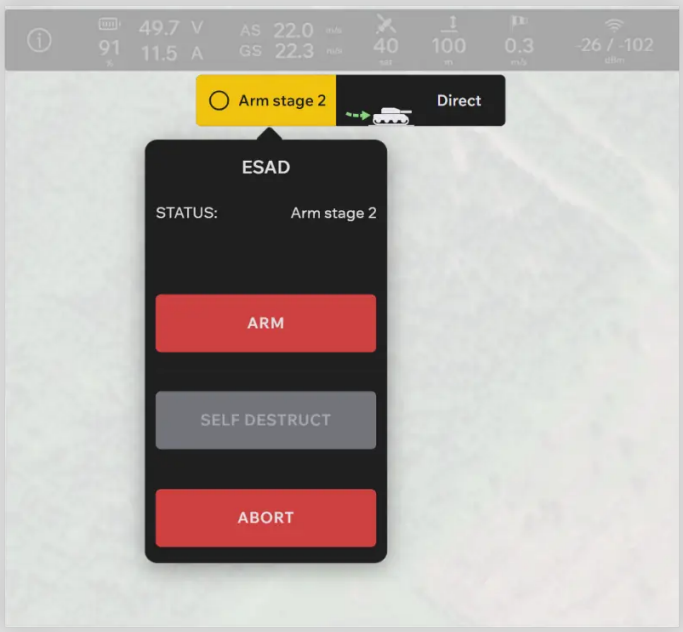


Figure 12.2: arming the warhead

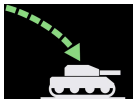
NOTE: If the UAV is no longer flying and the ESAD is **NOT ARMED**, the device will return back to **SAFE** mode, re-enabling all safety checks and discharging the initiator.

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12.2 Engagement Profiles

Avenger 5 has two operator selectable engagement profiles which change the trajectory flown by the UAV during the terminal guidance phase:

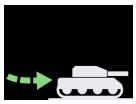
CAUTION: Due to more demanding flight profile, the engagement envelope is highly reduced when engaging moving targets from the front or sides.



Shallow – the UAV engages the target at a shallow trajectory, gradually increasing or decreasing the approach angle until the UAV is ~15° above the tracked target. This mode allows for the target to be hit at a shallow angle, or when low noise signature is required during the terminal phase of the engagement.

CAUTION: Take note of any terrain features or obstacles when using **Shallow** engagement profile as it may result in a collision or a loss of a valid track.

CAUTION: If initiating the engagement from a steeper angle than 15°, make sure that maximum dive airspeed is not exceeded.



Direct – the default mode, where the aircraft flies the most direct path towards the target, without any additional trajectory constraints. This mode has the widest engagement envelope and has the highest probability to hit the target from the front or side, but due to not having any control over its flight trajectory, the impact angle is purely dependent on the positioning of the UAV at the start of the engagement.

Currently selected mode icon is displayed in the **Main window**. To change the engagement profile:

1. Press **Engagement Profile** status icon.
2. Select the desired engagement profile.
3. Slide to confirm.

CAUTION: When UAV enters **Attack** mode, changing the engagement profile is no longer possible. If a different engagement profile is necessary, abort the attack and reengage with a new profile.

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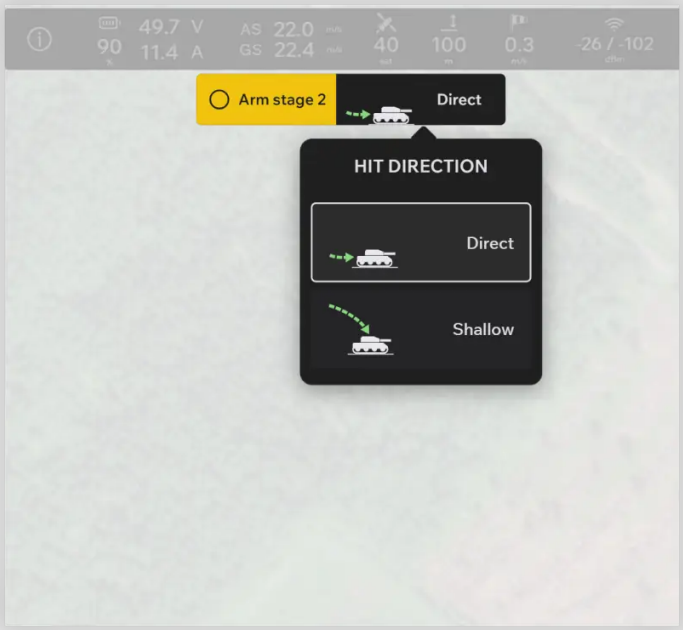


Figure 12.3: changing engagement profile

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12.3 Engagement

For the loitering munition to have the highest success rate it is important to assure that:

- **GCS Failsafe timeout** is set long enough, to prevent the UAV from entering **RTL** during the critical engagement phase.
- Warhead is **ARMED**.
- Target is tracked using the **Daylight (EO)** camera sensor.
- Target has a **valid track for 2-5 seconds** to allow for the targeting system to acquire the best possible estimates.
- Elevation towards the target is **no more than 15°** when starting the engagement.
- There are no environmental objects that will **obstruct** the target during the engagement phase.

CAUTION: Operator is responsible for assuring that all of these conditions are met before proceeding with target engagement.

WARNING: During the engagement, the UAV **MUST NOT** exceed 55m/s airspeed. If this airspeed is exceeded, the gimbal **MAY FAIL** to point towards the tracked target, and **will** result in missing the target. If this airspeed is exceeded, immediately **ABORT** the engagement and re-engage from a shallower angle.

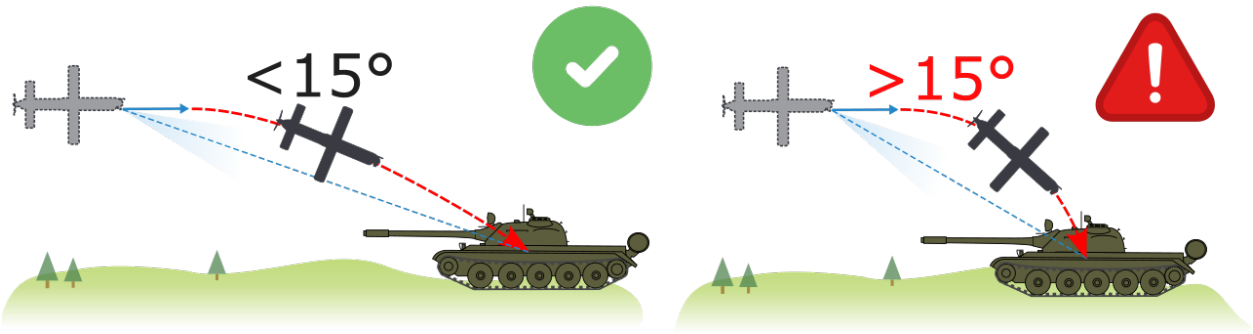


Figure 12.4: engagement angle

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To start the engagement:

1. Verify that the above mentioned conditions are met.
2. Press **Engage** button.
3. Slide to confirm.

When UAV enters **ATTACK** mode, it will acquire the target that is actively being tracked and turn towards the predicted lead angle required to hit the target. After matching the heading, the engagement controller will take over and the UAV will fly towards the tracked target, while compensating for target motion, wind, or manually moving the active track.

If the track is lost, the UAV will fly in dead-reckoning mode, based on the last known target elevation and heading.

CAUTION: The tracker **does not** have active object detection or dead-reckoning, which would re-track the target by itself if it was lost due to low light conditions or being obstructed by another object or environment. Operator **must manually re-track** or abort the engagement if the tracker loses the target.

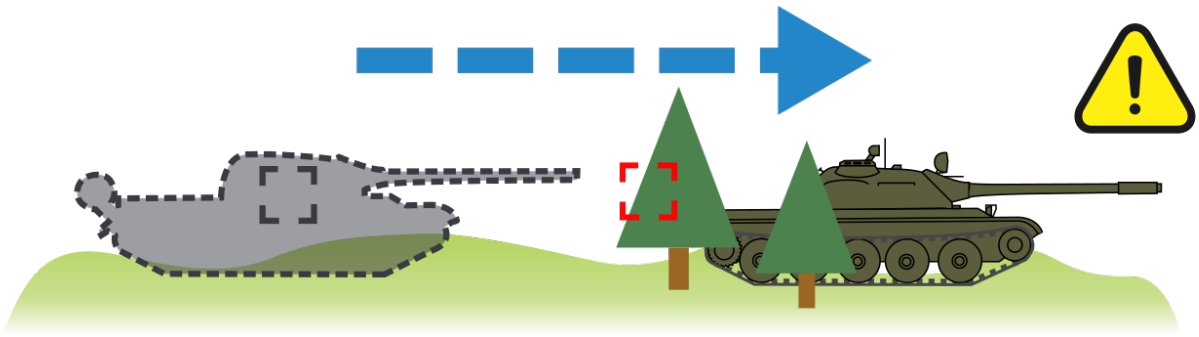


Figure 12.5: tracker behavior

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12.4 Engaging Abort

When UAV engages a target in **ATTACK** mode, a slider bar will appear at the bottom of the main view screen with the message **Abort Engaging**.

To abort engagement, simply slide the **Abort Engaging** bar to the right.

When abort is initiated:

1. UAV will lock its current heading and immediately initiate a climb towards **Takeoff** altitude.
2. After reaching the **Takeoff Altitude** UAV will enter **GUIDED** mode and start a standard loiter around the abort location at **Abort Engaging** altitude.
3. If no radio link is present, UAV will enter **RTL** and start flying towards the **Return Point**.

WARNING: If **Attack** abort was initiated after the estimated time for safe abort has expired. The UAV would still wave-off from the target, but has a high probability of crashing into nearby objects or terrain.

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13 Operation Scenarios

13.1 GNSS Denied Operation

When operating in a GNSS denied environment, **SwarmC2** software supports required tools to operate all UDS UAV systems in such an environment.

CAUTION: GNSS denied operation requires elevated workload from both **Payload Operator** and **Pilot** and poses additional risk of losing the UAV. Read the UAV and GCS manuals carefully and familiarize with vehicle behavior, available functionality and best practices, before UDS UAVs in GNSS denied environment.

13.1.1 Takeoff Without GNSS

If no GNSS is available on the ground, UAV requires to manually enter its initial position and wind estimate before takeoff.

CAUTION: Wind estimate will always default to 0m/s, and if no value is entered, vehicle may experience very large drift from its true position. Make sure to enter wind speed and direction to your best knowledge before takeoff to minimize risk of losing the UAV.

13.1.2 UAV Location Correction in GNSS Denied Environments

When operating in GNSS-denied environments, if camera and vehicle locations on the map differ heavily from what is seen by the payload—but the UAV's or landmark's actual location is known to the operator—you can manually correct its position on the map using the **I'm Here**, or **I'm Looking Here** functionality.

Correct the position as follows:

1. Tap a point on the map that's supposed to be the correct UAV's location.
2. Tap **"I'm Here"** or **"I'm Looking Here"**, depending on what the selected point represents.
3. Slide to confirm.
4. After confirmation, the UAV's position on the map will be updated, the flight path will automatically adjust, and wind speed and direction estimates will be recalculated to ensure accurate navigation to the next waypoint.

NOTE: The **I'm Here** button is only available when the UAV is not using GNSS for navigation.

13.1.3 GNSS Disabling in Spoofing Conditions

NOTE: Disabling GNSS is only recommended when signs of spoofing are observed.

CAUTION: Disabling GNSS during normal flight conditions reduces flight accuracy and may result in UAV damage or harm to nearby areas.

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If spoofing is suspected and the UAV begins deviating from its planned flight path, you can disable the GNSS system as follows:

1. On the **Flight Toolbar**, tap on the GNSS information icon.
2. In the **GPS Status** window, tap Disable **GNSS**.

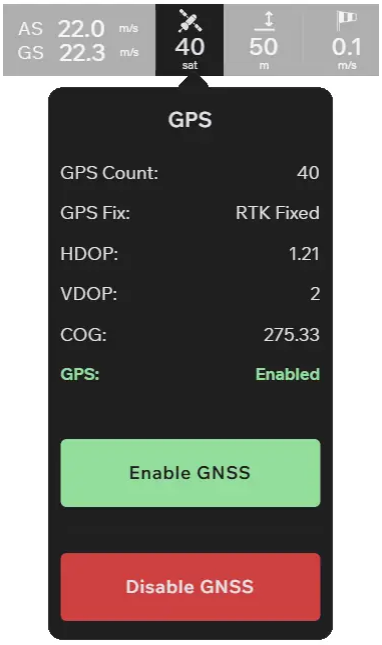


Figure 13.1: Disable GNSS

3. Once disabled, GNSS icon will turn red. This indicates that the UAV is now operating on inertial navigation only.

WARNING: When flying without GNSS make sure to regularly update vehicle position using known landmarks using **I'm Here** and **I'm Looking Here** functionality, as inertial navigation will continually drift from true position without GNSS.

CAUTION: If radio communication is lost while GNSS is disabled, the system will automatically re-enable GNSS after a certain amount of time, specified by **GPS Failsafe Timeout** parameter configured from flight safety options menu.

4. To enable the GNSS system back, tap the Enable GNSS button in the GNSS information window. Once GNSS is re-enabled, the warning message will disappear, and the UAV will start using GNSS for navigation if available.

CAUTION: If GNSS was disabled during flight, it will remain disabled unless a radio GNSS failsafe is triggered. Make sure to re-enable it before entering a safe area to avoid the risk of losing the UAV.

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14 Terminology Glossary

This glossary defines common terms and abbreviations used throughout the **SwarmC2** manual.

14.1 Core System Terms

Term	Description
UDS	Unmanned Defense Systems; manufacturer of the UAV platforms supported by SwarmC2.
SwarmC2	Ground control software used to operate supported UDS UAV systems.
GCS	Ground Control Station; operator station used to control UAVs, monitor telemetry, and manage mission execution.
UAV	Unmanned Aerial Vehicle; aircraft controlled either autonomously through mission plans or manually by operators.
Forecaster	UDS fixed-wing ISTAR UAV platform
Avenger	UDS Loitering Munition platform
Payload	Mission equipment carried by the UAV, typically EO/IR cameras and related control systems.
GCS Operator	Operator responsible for flight setup, mission control, and in-flight UAV operation from SwarmC2.
Payload Operator	Operator responsible for camera/payload use, observation, and target-related actions.
Launch Assistant	Crew member responsible for safe physical launch and post-landing handling tasks.

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14.2 Navigation and Positioning

Term	Description
GNSS	Global Navigation Satellite System; satellite-based positioning used for navigation (for example GPS/Galileo).
GNSS-denied	Operational condition where GNSS is unavailable, unreliable, jammed, or intentionally disabled.
MSL	Mean Sea Level; altitude reference measured relative to average sea level.
AGL	Above Ground Level; altitude measured relative to the terrain directly below the aircraft, distinct from MSL.
MGRS	Military Grid Reference System; grid-based coordinate format used for tactical location reporting.
Geofence	Configured geographic boundary that constrains or governs UAV flight behavior.
Rally point	Predefined alternate safe location uploaded with the mission plan; the autopilot may use it during RTL instead of the home or return point when it is nearer and within configured limits.
POI	Point of Interest; user-created map marker used for reference during planning or execution.

14.3 Flight Modes and Behaviors

Term	Description
AUTO	Automatic mission execution mode using uploaded waypoints and mission logic.
CRUISE	Stabilized flight mode optimized for direct heading/altitude style operator control.
LOITER	Mode/behavior where UAV circles or holds around a selected point.
GUIDED	Map-commanded flight mode activated when the operator issues a Loiter Here or map-pattern command; not selectable directly from the Flight Mode dropdown.
RTL	Return To Launch; failsafe or commanded behavior to return toward launch/home according to configured safety options.
Home point	The recorded launch/takeoff location that the UAV navigates toward during RTL.
Go-around	Landing abort behavior where UAV exits landing and repositions for another landing attempt.
Deep Stall	Landing technique where the UAV transitions to a near-vertical descent to land in a confined area; the basis for the Quick Deep Stall profile.
Quick Deep Stall	Landing profile variation that transitions to deep-stall landing from a specific approach geometry.

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14.4 Payload and Video Terms

Term	Description
EO	Electro-Optical; daylight/visible-spectrum camera feed.
IR	Infrared; thermal camera feed used for low-visibility or heat-based observation.
FOV	Field of View; angular extent of the camera's visible scene, displayed in the payload HUD overlay.
NUC	Non-Uniformity Correction; thermal camera recalibration to compensate for temperature-related sensor drift.
STOW	Camera/gimbal retracted position; used during takeoff, landing, and when the payload is not in use.
NADIR	Camera mode where the gimbal points straight down toward the ground.
OBS	Camera mode where the view follows the aircraft platform's motion (body-referenced).
PTC	Stabilizes the camera on map coordinates after you select a map point and choose **Observe** from the function menu.
GRR	Payload camera mode where joystick movement controls the view, then stabilizes on map coordinates when joystick input stops.
Multitrack	Payload capability for maintaining and managing multiple tracked objects (if available on current payload build).

14.5 Reporting and Tactical Map Functions

Term	Description
5W	Five W report; structured tactical report marker used on the map for shared situational awareness.
BDA	Battle Damage Assessment; post-strike/effects assessment update attached to a tactical report.
CFF	Call For Fire; formatted fire-support request generated from selected map coordinates.
Set Home Position	Function to set the selected map point as the UAV's default RTL destination.
"I'm Here"	Function to correct UAV map position in GNSS-denied operation using known true location.
"I'm Looking Here"	Function to correct navigation estimate based on known viewed point and payload geometry.

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14.6 Connectivity and Integration

Term	Description
TAK	Team Awareness Kit; tactical situational awareness ecosystem used for shared map data and collaboration.
ATAK	Android Team Awareness Kit; Android client in the TAK ecosystem.
SitaWare	External tactical awareness software that can exchange operational data with SwarmC2.
Uplink	Radio link from the GCS to the UAV; carries commands and control data.
Downlink	Radio link from the UAV to the GCS; carries telemetry and video data.
RSSI	Received Signal Strength Indicator; measure of radio link signal strength used for link quality monitoring.
MBTiles	Offline map tile file format; loaded via General Settings to enable map display without network access.

14.7 Hardware and Safety Terms

Term	Description
RBF pin	Remove Before Flight; safety pin used in pre/post-flight handling procedures.
Rotator	Directional antenna positioning mechanism used to improve radio link performance.
Auto Follow (Rotator)	Rotator mode that automatically points toward the currently selected UAV.

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